ECO 413: OPERATIONS RESEARCH
LECTURE 1

What is Operations Research?

Operations Research (or Operational Research in British English) often shortened to *OR*, is an analytical method of problem-solving and decision-making that is useful in the management of organisations. In operations research, problems are broken down into basic components and then solved in defined steps by mathematical analysis. It is considered to be a subfield of Mathematics. The term management science is occasionally used as a synonym. Because of its emphasis on practical applications, operations research has overlap with many disciplines. Originating in military efforts during World War II (1939 – 1945), its techniques have grown to concern problems in a variety of industries.

OR is often concerned with determining the extreme values of some real-world objective: the maximum (of profit, performance, or yield) or minimum (of loss, risk, or cost). It is used to solve the problems that often challenge business owners and managers. It encourages businesspeople to use advanced analytical tools to make more informed and effective decisions for their organisations. Employing different techniques, OR arrives at optimal or near-optimal solutions to decision - making problems. For example, in order for *Uber* to have a master routing plan, it has to decide which driver should be sent where, when, and how much they should charge the customers.

Types of Mathematical Models used in OR

Important mathematical models often used in OR include *Allocation or Optimization Models* used in optmizing a given objective such as profit maximization under certain restrictions; Scheduling Models used in determining an optimum sequence of performing ceertain operations or jobs with a view to minimizing overall time and/or cost; *Queuing Models* employed to

minimize the cost of service and waiting in a service facility; *Inventory Models* used to optiize invetory levels to minimize costs associated with inventories; *Replacement Models* which are concerned with machine and equipment replacement strategies; and *Simulation Models*.
The different types of mathematical models we will consider in this course can be categorised into two:

- **Deterministic models**- these are models that attempt to establish exact mathematical relationships between the variables concerned, e.g., LP, Transportation Problem.

- **Stochastic Models**- these are models that use probabilities to describe average rather than precise behaviour of the system, e.g., simulation models, queues, replacement , maintenance, and reliability models.

The Mathematical Modelling Process

i. **Abstraction**: This is the first step in modelling and it involves selecting the critical factors or variables from an empirical situation. The objectives of the model are important as they affect the choice of the variables of interest. For example, two people can observe the same system, say a bank. The first person may be interested in the patterns of customer arrival and departure, while the second person may be interest in fuel consumption and machine maitenance. Each person will develop a completely different model of the bank because the objectives and the quantities they measure are not the same.

ii. **Model Building**: This is the second step in modelling. The relevant factors or variables slected durign abstraction are put in some logocal manner so that they form a model of the given problem.

iii. **Deriving the Solution**: Some mathematical calculations are often required for obtaining the solution to the model that has been developed.

iv. **The Question of Error**: There is always some degree of error in the abstraction process. If the error involved is not big enough to disturb the validity of the of the model, it is considered acceptable. When the error is significant, the model has to be re-built.

v. **Updating the Model**: A mathematical model should be modified as soon as one or more of the controlled variables change significantly.

Advantages of Modelling

1. Models can be manipulated in a variety of ways until the best solution of te problem under consideration is obtainted. The actual system can then be operated in an optimum fashion with a minimum of costly experiments or adjustments.
2. Models can be used on a ‘what if’ basis to explore possible future consquences of decisions that might be made today.
3. Modelling provides an easy and systematic appraoch for studying the problem concerned.
4. Modelling provides an opportunity to explore a broad range of solutions and to estimate the sensitivity of the solution to changes in individaul elements or in combinations of these elements.

Methods of Analysis

Mathematical models are analysed by a variety of methods.The ones we will consider are:

- The use of formulas
- The use of algorithms, i.e., step by step procedures which are used when formulas will not work
- Simulation- this method is used where there are no appropriate formulas or algorithms available for the analysis of a model.

A Word of Caution about Modelling

It should be noted that a mathematical model is only useful when the model is a reasonable approximation of reality. If the model is a good representation of the problem, the solution of the model will be a good solution of the problem, but the solution of a poor model even though it may be an exact one will not be a good solution of the problem. Hence, the construction of the model and the translation of the results to the actual problem must be done with utmost care. The problem must be well defined before constructing the model. We must construct an appropriate model of a problem if we can, otherwise we must use an intuitive or heuristic approach to solve the given problem.

ECO 413: OPERATIONS RESEARCH
LECTURE 2

OPTIMIZATION MODELS

1. **Linear Programming (LP)**

This is a structured method of solving problems in which the objective function and constraints are expressed as linear mathematical statements and the decision variables have continuous values. It is widely used for solving production and operations management problems associated with allocation of limited resources to competing activities in an optimal manner. Thes types of problems are handled using ***Optimization Models*.** Engineers use optimization approach to allocate resources in the best way possible to satisfy acceptable performance standards, e.g.,

- Product mix
- Acceptable production levels
- Assignment of people to tasks
- Similar problems dealing with limited time, money, workforce, and technology

The 'optimum result' could be maximum benefits, minimum costs, highest effectiveness of technical personnel, or the most efficient engineering design

Key terms used in LP formulations

- **Decision Variables**: These are the unknowns of the LP problem. Their values represent decisions such as the number of hours each engineer should spend on each task, the amount of raw materials to be used in a blending process, or the quantity of each product to be manufactured in a plant.

- **The Cost Vector**: The unit contribution of the decision variables to the total measure of effectiveness of the system which may be total cost, total profit, etc. For example, if the total cost of the system is the measure of effectiveness, the hourly salaries of engineering personnel, the unit costs of raw materials or the unit production costs of the final product constitute the cost vector; if total profit is taken as the measure of effectiveness, unit profits, rather than the costs, are used in the cost vector. The unit contributions, i.e., costs, profits, opportunity costs, etc, can take on positive, negative, integer, or non-integer values to be estimated before the model is built

- **The Objective Function:** The linear combination of the individual contributions of the decision variables to the total measure of effectiveness of the system. The LP problem is formulated to obtain either the maximum or the minimum value of the objective function

- **The Constraints**: The linear expressions that describe the interactions among the decision variables and define the functional relationships of the decision system. Each constraint specifies a resource limitation, a certain technical requirement, or a demand imposed upon the system. For m number of such constraints, the coefficients associated with each of the n decision variables in each of the m constraints constitute an n x m matrix. Constraints could be expressed as being less than, equal to, or greater than a constant

The Assumptions Underlying LP Models

i. Finite number of choices are available to a decision maker and the decision variables do not assume negative values (i.e., they are either zero or positive). This is otherwise refered to as the *Finite Choices Assumption*.
ii. Objective function is a linear function of the decision variables i.e., there are constant returns to scale- if one init of a product contributes ₦5 towards profict, then 2 units will contribute ₦10. This is otherwise refered to as the *Proportionality Assumption*.
iii. Constraints can be expressed as linear equalities or inequalities
iv. Unit contributions of decision variables, resource levels, and coefficients of decision variables in the constraints all have deterministic values. This is otherwise refered to as the *Certainty Assumption* which implies the prior knowledge of all the coefficients in the objective function, the constraints and the resource levels.
v. Decision variables can take on continuous (non-integer) values. This is otherwise refered to as the *Continuity Assumption*.
vi. The total of all the activities is given by the sum total of each activity conducted separately, i.e., the total profit in the objective function is the sum of theprofit contributed by each of the products separately. This is otherwise refered to as the *Additivity Assumption*.

Points to Note

- The name LP stems from assumptions 2 and 3
- Assumptions 4 and 5 hold true for the ordinary LP problems, but they can be released in extensions of LP
- Assumption 1 can also be released by modifying the basic LP model to handle non-positive (zero or negative) or un-restricted (zero, positive or negative) variables if such variables exist in the decision system

LP Model Formulation

Minimize $Z = \sum_{i=1}^{n} (X_i)(C_i)$

Or

Maximize $Z = \sum_{i=1}^{n} (X_i)(C_i)$

Subject to:

Constraint 1

Constraint 2, et c

Where Z = value of the objective function

C_1 --- C_n = Unit contribution of the decision variables to the total measure of effectiveness of the system

X_1 --- X_n = Optimum number of each the decision variables required to maximize or minimize the objective function

For example, if the constraint is the total time available for production T, and if the coefficient t_i is the time taken to perform function i, the constraint takes the following form:

$$\sum_{i=1}^{n} (X_i)(t_i) \subseteq T$$

X_i --- $X_n \supseteq 0$ = Non-negativity constraint

Criteria for Using LP

- Problem involves distribution of limited resources among competing demands
- Problem can be described in terms of numerical variables with a well-defined objective function (e.g., maximization of profit or minimization of cost)
- A set of interdependent linear, non-negative variables can be generated with alternative solutions

Points to Note

- LP models are formulated in three steps:
  - Identification of the decision variables
  - Development of the objective function as a linear function of the decision variables
  - Mathematical definition of the constraints
- Measures of effectiveness and time horizons vary from one project to another

Example 1

Suppose you have two Processing Machines (1 and 2) with which you intend to process two types of products (A and B). Machine 1 requires 8 hours and 4 hours to process products A and B respectively, compared to 4 and 8 hours respectively required by Machine 2 to process both products. The total time Machines 1 and 2 are available for production are 120 hours and 96 hours respectively. The unit contribution of products A and B to profit are ₦180 and ₦150 respectively. Formulate the LP problem. Determine the optimal combination of the two products

LP Formulation

Maximize P = 180A +150B (Objective Function)
Subject to:
8A + 4B = 120 (Time constraint in Centre 1)
4A + 8B = 96 (Time constraint in Centre 2)
A,B >= 0 (Non-negativity constraint)

Example 2

Suppose that, as Project Manager, you are planning for the production of two products (X_1 and X_2) for the coming week. It has been determined that each unit of product X_1 contributes 300 Naira to profit while each unit of product X_2 contributes 400 Naira; each unit of X_1 requires 4 man-hours and 2 Kg of materials, while each unit of X_2 requires 2 man-hours and 3 Kg of materials. The total quantity of material available for the week is 125, while the available man hour for the week is 100. Formulate this information as an LP problem and solve the it.

LP Formulation

Maximize P= $300x_1 + 400x_2$ (Objective Function)
Subject to:
$4x_1 + 2x_2 = 100$ (Man hour constraint)
$2x_1 + 3x_2 = 125$ (Material constraint)
$x_1, x_2 >= 0$ (Non-negativity constraint)

Example 3

A factory requests you to supply on a daily basis a minimum of 6 units of products of grade I, 6 units of grade II, and 5 units of grade III. Assume that you service the factory from two sources, A and B; and that 2,1, and 1 units of grades I, II and III respectively are delivered per trip from source A, while 1, 2, and 1 units of grades I, II and III respectively are delivered per trip from source B. Given that x and y daily trips are made from A and B respectively and each trip costs N5000 and N8000, formulate the LP problem and determine the minimum cost of transporting the units of products to the location.

LP Formulation

Assume x = daily trips from Source A, y= daily trips from Source B
Min C = $5000x + 8000y$ (Objective Function)
Subject to:
$2x + y >= 6$ (Grade I supply constraint)
$x + 2y >= 6$ (Grade II supply constraint)
$x + y >= 5$ (Grade III supply constraint)
$x, y >= 0$ (Non-negativity constraint)

Exercise

An integrated wood products factory produces plywood and particleboard. The time to produce a unit of each product, the monthly capacity of operations and the net profit per unit of each product are as given in Table 1. Formulate the LP problem.

Table 1: Product Manufacturing Information

Operation	**Time per unit (hours)**		**Operation Capacity (hrs/month)**
	Plywood	**Particleboard**	
1	1	2	80
2	8	3	480
3	5	4	400
Net Profit per Unit (₦)	1350	1500	

PRIMA AND DUAL LP PROBLEMS

- The original LP problem is the PRIMA

- From the PRIMA, we can formulate the DUAL; i.e., from a PRIMA minimization LP problem, we can formulate a DUAL maximization problem and vice-versa

Solution Procedure

1. Convert the coefficients of the decision variables in the objective function of the PRIMA to the constraint limits of the DUAL
2. Convert the set of coefficients of one decision variable in all constraints of the PRIMA to a set of coefficients of decision variables in the DUAL. Repeat the process for all other decision variables
3. Convert the constraint limits in the PRIMA to the respective coefficients of the decision variables in the DUAL to produce the objective function of the DUAL

Duality in Matrix Format

Max $Z = CX$
Subject to $AX \leq B$
$X \geq 0$

***Is equivalent to*:**

Min $Z^1 = B^T Y$
Subject to $A^T Y \leq C^T$
$Y \geq 0$
Where A^T = Transpose of Matrix A
B^T = Transpose of Matrix B

Example

If the PRIMA is:
Minimize $C = 5000x + 8000y$
Subject to:
$2x + y >= 6$
$x + 2y >= 6$
$x + y >= 5$
$X, Y >= 0$
Construct the DUAL from the PRIMA

Solution

The DUAL of the PRIMA is:
Maximize $P = 6z_1 + 6z_2 + 5z_3$
Subject to:
$2z_1 + z_2 + z_3 <= 5000$
$z_1 + 2z_2 + z_3 <= 8000$
$z_1, z_2, z_3 >= 0$

Exercises

Construct the DUAL from the PRIMA of examples 1 to 4 and solve the DUAL LP problems

LP Solution Techniques

There are several techniques for solving LP problems. The three that will be demonstrated in this course are:

- Graphical method
- Simplex method (Most popular and the basis of most computer solutions, but rather tedious if done manually. There are online programmed solution techniques)
- Use of different computer software – *EXCEL* SOLVER, Matrix Pivot Tool, LINGO, et c

Steps Involved in Using the Graphical Method

This method can be used only for 2 decision variable problems with each product plotted on each axis. The procedure is as follows::

1. Draw the graphs of the Constraint Equations.
2. Identify the region common to the graphs,commonly refered to as the'solution space'. Observe that this region is a closed polygon for a maximization problem, (i.e., the constraint for a maximization problem is typically $\leq$), while it is an open space bounded by straight lines for a minimization problem (i.e., the constraint for a minimization problem is typically $\geq$).
3. Identify the joints of the region as the points of feasible solution of the LP problem.
4. Substitute each of the coordinates of the joints into the objective function. The values of the objective function at these points are the 'feasible solutions'of the LP problem. For a maximization problem, thehighest of these values is the optimum solution of the LP problem. For a minimization problem, the smallest of these values is the optimum solution of the LP problem.

Example 4

Solve the following LP problem graphically.

Maximize P = 400x + 500y

Subject to:

$x + 2y \leq 9$

$x + y \leq 7$

$y \leq 3$

Solution

Consider the line x + 2y = 9

When x = 0, y = 4.5; (0, 4.5) satisfies the line; when y = 0, x= 9; (9, 0) satisfies the line

Next: Consider the line x +y = 7

When x = 0, y = 7; (0, 7) satisfies the line; when y = 0, x= 7; (7, 0) satisfies the line

The graphs of the constraints are shown in Figure 1. The solution space is the shaded region ABCDO. The points of feasible solution A(0,3), B(3,3), C(5,2), D(7,0),and (0,0)

Recall that the Objective Function is P = 400x + 500y

At A(0,3), P = 400x0 +500x3 = 1500

At B(3,3), P = 400x3 +500x3 = 2700

At C(5,2), P = 400x5 +500x2 = 3000

At D(7,0), P = 400x7 +500x0 = 2800

At (0,0), P = 400x0 +500x0 = 0

Maximum Profit, P= ₦3000 is realised at C(5,2)

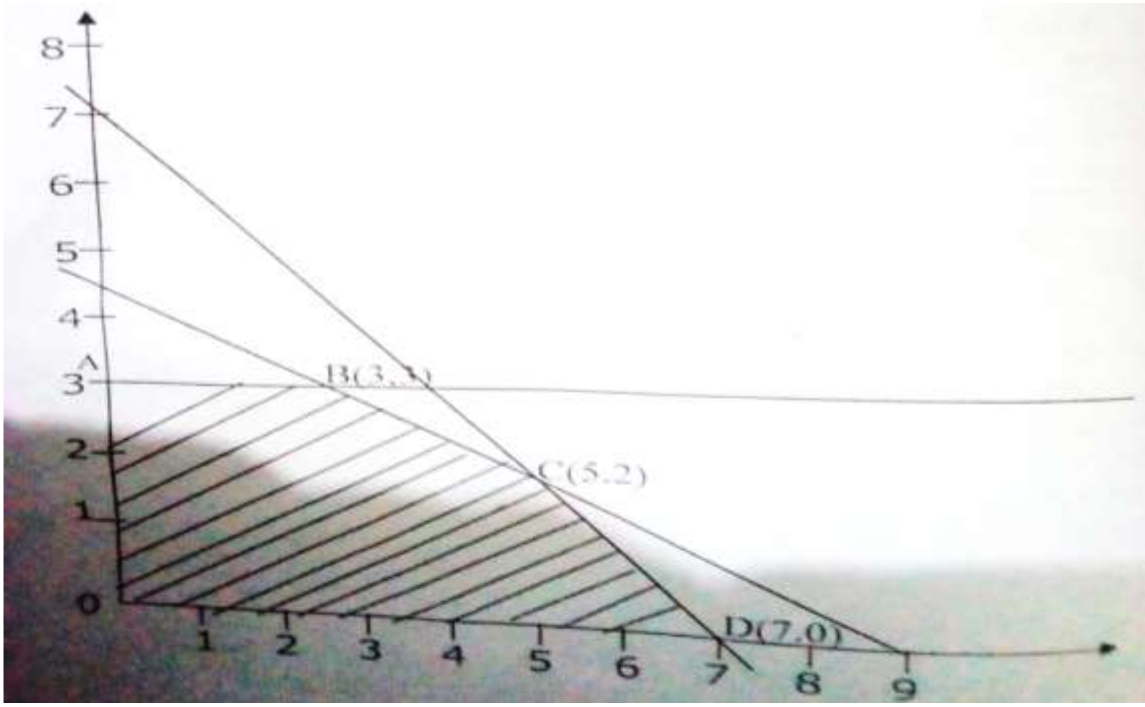


Figure 1: Graphical Solution

The Simplex method

The Simplex method is an algebraic procedure that, through a series of repetitive operations, progressively approaches an optimal solution. Theoretically, the Simplex can solve a problem consisting if any number of variables and constraints althogh for problems containing more than four variables or four constraints, the actual calculations ae best left to the computer. There are many LP softwarepackages available. *The Simplex method will find the optimal solution to any LP problem in a finite number of iterations. The expected number of iterations required by the Simplex methid has been estimated as twice the number of inequality constraints/number of constraint equations.*

Steps Involved in Using the Simplex method

1. Re-write the Constraint Equations in stadard form. This is done by turning each of the inequalities into an equation. In doing this, a slack variable is added to each inequality.
2. Arrange the stadard constraint equations and the Objective Function in a table called *Initial Simplex Tableau*
3. Select a ***Pivotal Column*** by determining the column with the most positive entry in the Objective Function row. If there are several candidates for a pivotal column, choose any one arbitrarily
4. Select a ***Pivotal Row***- Form the ratios of the entries in the rightmost column by the corresponding entries of the pivotal column for those entries in the pivotal column that are positive. *The pivotal row is the row for which the smallest of these ratios occur*. If

there is a tie, choose any one of the qualifying rows. If none of the entries in the pivotal column is positive, the problem has no finite optimum solution. Stop computation.

5. Identify and circle the ***Pivot Element***, i.e., entry at the ***intersection*** of the ***Pivot Element*** and the ***Pivot Row***
6. Divide **ALL** the values in the ***Pivot Row*** by the ***Pivot Element***
7. Perform ***Repetitive Row-by-Row Operation*** **otherwise called Pivotal Elimination** by subtracting the appropriate multiples of the pivotal row from all other rows including the objective function row so that all elements in the pivotal column *except the pivotal* element become zero.
8. Construct a new tableau. In the new table, replace the label on the pivotal row by the entering variable
9. Apply the optimality test to check if there is any positive value greater than zero in the Objective Function row.
10. If there is any one, go back to Step 3 to select a new pivot, If there is none , you have obtained the solution

It should be noted that:

- In general, a LP problem may have: no feasible solution, i.e., there are no ponts satisfying all the constraints; a unique optimal solution; or more than one optimal solution.
- A point satisfying all the constraints of a LP problem is called feasible solution.; the set of all such points is the feasible region. It is not every LP problem that has a feasible region.
- Slack variable is so called because it takes up the slack between the two sides of the inequality.
- When the PRIMA Problem is solved by the Simplex Method, the final Tableau contains the optimal solution to the DUAL problem in the Objective Row under the columns of the slack variables.

Worked Example 1

LP Problem Formulation

Maximize $p = 135x + 90y$

subject to:

$2x + y \leq 40$

$3x + 8y \leq 240$

$4x + 5y \leq 200$

$x, y \geq 0$

Convert the Equations into Standard Form by adding SLACK Variables (Note that the number of slack variables required = Number of Constraint Equations, i.e., if there are two constraint equations there will be two slack variables. In this particular example, there are three constraint equations, hence, three slack variables, z_1, z_2, z_3 are required).

The Constraint Equations in Standard Form is as Follows:

$2x + y + z_1 + 0z_2 + 0z_3 = 40$

$3x + 8y + 0z_1 + z_2 + 0z_3 = 240$

$4x + 5y + 0z_1 + 0z_2 + z_3 = 200$

$p = 135x + 90y0z_1 + 0z_2 + 0z_3$

Set Up the Initial Simplex Tableau: The initial Simplex Tableau is set up as follows, with the pivotal column indicated:

Solution Variable	Products		Slack Variables			Solution Quantities
	x	**y**	z_1	z_2	z_3	
z_1	2	1	1	0	0	40
z_2	3	8	0	1	0	240
z_3	4	5	0	0	1	200
p	**135**	90	0	0	0	0

Pivotal Column

Solution Variable	Products		Slack Variables			Solution Quantities	Solution Quantities Divided by Elements in Pivotal column
	x	**y**	z_1	z_2	z_3		
z_1	**2**	**1**	**1**	**0**	**0**	**40**	**40/2 = 20**
z_2	3	8	0	1	0	240	240/3 = 80
z_3	4	5	0	0	1	200	200/4 = 50
p	**135**	90	0	0	0	0	

Pivotal Row

Pivotal Element

Second Simplex Tableau (Produced by dividing all entries in the pivotal row by the pivotal element which is 2)

Solution Variable	Products		Slack Variables			Solution Quantities
	x	**y**	z_1	z_2	z_3	
x	1	0.5	0.5	0	0	20
z_2	3	8	0	1	0	240
z_3	4	5	0	0	1	200
p	**135**	90	0	0	0	0

Third Simplex Tableau (Produced by pivotal elimination)

	Solution Variable	Products		Slack Variables			Solution Quantities
		x	**y**	z_1	z_2	z_3	
Pivotal Row 1	x	1	**0.5**	0.5	0	0	20
Row 2 – 3Row 1	z_2	0	**6.5**	-1.5	1	0	180
Row 3- 4 Row 1	z_3	0	**3**	-2	0	1	120
Row 4 -135 Row 1	p	0	**22.5**	-67.5	0	0	-2700

New Pivotal Column

Determining the Pivotal Row and Pivotal Element

Solution Variable	Products		Slack Variables			Solution Quantities	Solution Quantities Divided by Elements in Pivotal column
	x	**y**	z_1	z_2	z_3		
x	1	0.5	0.5	0	0	20	20/20.5= 40
z_2	**0**	**6.5**	**-1.5**	**1**	**0**	**180**	**180/6.5 = 22.7**
z_3	0	3	-2	0	1	120	120/3 = 40
p	o	22.5	-67.5	0	0	-2700	

Pivotal Row

New Pivotal Element

Fourth Simplex Tableau (Produced by dividing all entries in the pivotal row by the pivotal element which is 6.5)

Solution Variable	Products		Slack Variables			Solution Quantities
	x	**y**	z_1	z_2	z_3	
x	1	0.5	0.5	0	0	20
y	0	1	-0.23	0.15	0	27.7
z_3	0	3	-2	0	1	120
p	0	22.5	-67.5	0	0	-2700

Fifth Simplex Tableau (Produced by pivotal elimination)

	Solution Variable	Products		Slack Variables			Solution Quantities
		x	**y**	z_1	z_2	z_3	
Row 1-0.5 Row 2	x	1	0	0.62	-0.08	0	6.15
Pivot Row 2	y	0	1	-0.23	0.15	0	22.7
Row 3-3Row2	z_3	0	0	-1.31	-0.45	1	36.9
Row 4 -22.5Row 2	p	0	0	-62.33	-3.38	0	-3323.25

Hence the LP Problem solution is: x = 6.15; y = 22.7; p = ₦3323.25 (ignore the negative sign)

Worked Example 2

Maximize $p = (1/2)x + 3y + z + 4w$

subject to:

$x + y + z + w \leq 40$

$2x + y - z - w \geq 10$

$w - y \geq 10$

Simplex Solution

Tableau #1

x	y	z	w	s1	s2	s3	p	
1	1	1	1	1	0	0	0	40
2	1	-1	-1	0	-1	0	0	10
0	-1	0	1	0	0	-1	0	10
-0.5	-3	-1	-4	0	0	0	1	0

Tableau #2

x	y	z	w	s1	s2	s3	p	
0	0.5	1.5	1.5	1	0.5	0	0	35
1	0.5	-0.5	-0.5	0	-0.5	0	0	5
0	-1	0	1	0	0	-1	0	10
0	-2.75	-1.25	-4.25	0	-0.25	0	1	2.5

Tableau #3

x	y	z	w	s1	s2	s3	p	
0	2	1.5	0	1	0.5	1.5	0	20
1	0	-0.5	0	0	-0.5	-0.5	0	10
0	-1	0	1	0	0	-1	0	10
0	-7	-1.25	0	0	-0.25	-4.25	1	45

Tableau #4

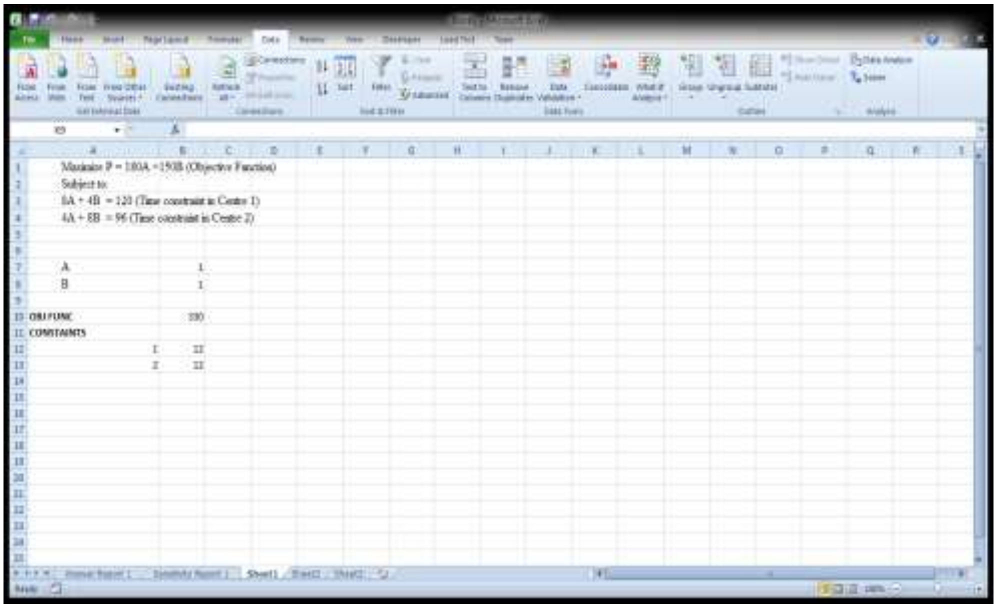
x	y	z	w	s1	s2	s3	p	
0	1	0.75	0	0.5	0.25	0.75	0	10
1	0	-0.5	0	0	-0.5	-0.5	0	10
0	0	0.75	1	0.5	0.25	-0.25	0	20
0	0	4	0	3.5	1.5	1	1	115

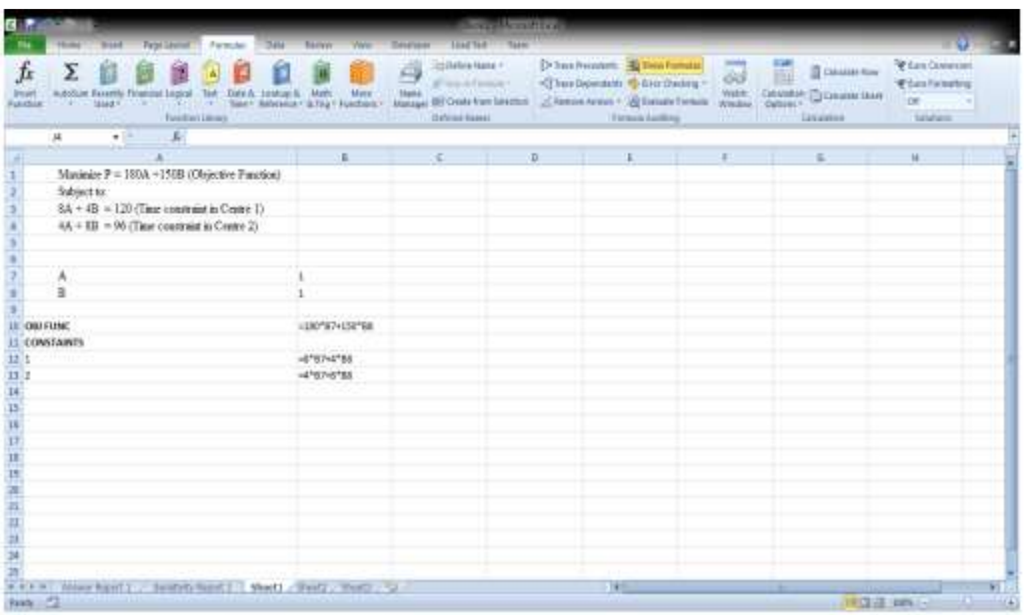
Sample of an Online Solution Simplex Solution Software

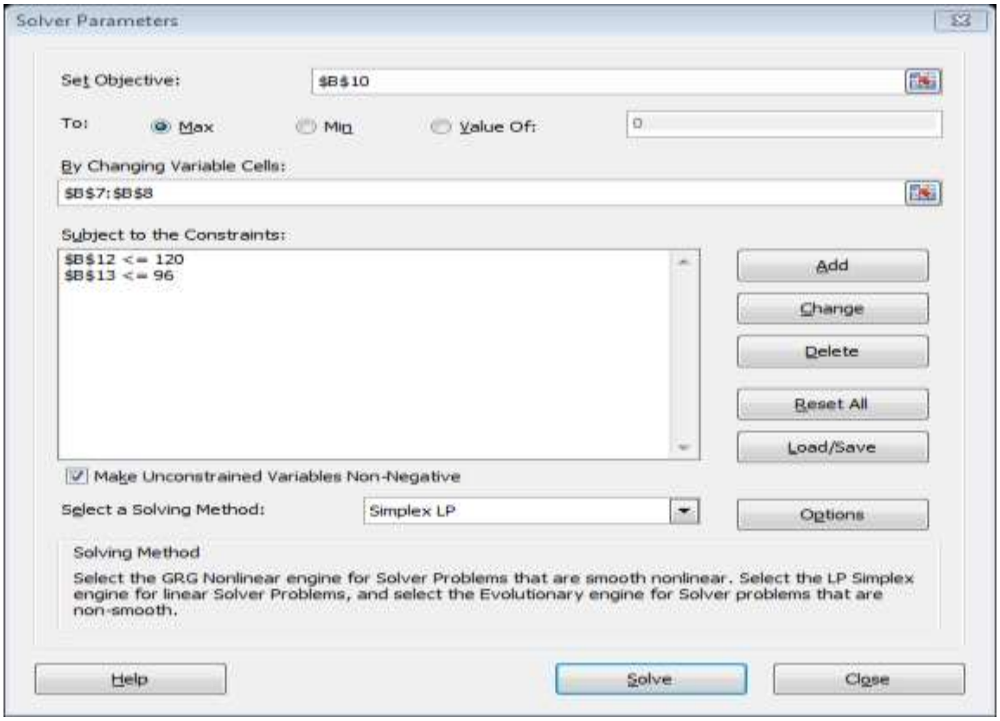
Steps Involved in Using *MS Excel Solver*

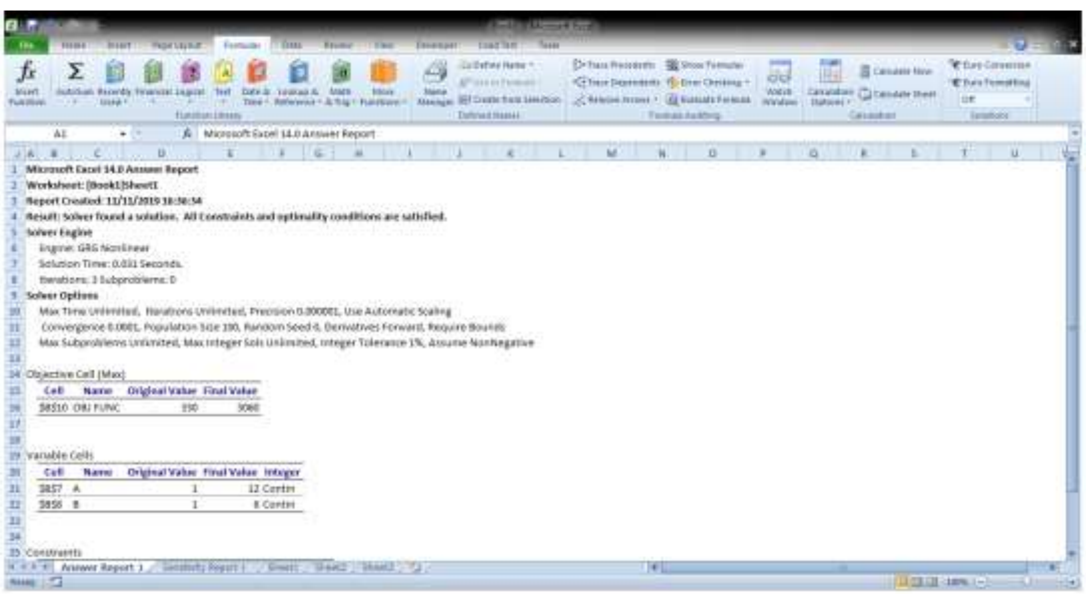
1. Open a new excel worksheet
2. List the variables in a column of the spread sheet
3. Initialize the variables by inputting a dummy value of 1 as the initial value of each of the variables. List in the column next to the first column.
4. Use cell references of the variables to solve the objective function equation with the dummy values of the decision variables.
5. Use cell references to solve the constraint equations listed in order in rows
6. Click on 'Data' and then click on 'solver;' (if it is already installed on your computer. If not, on the menu bar, Click on ***File***; Click on, '***Options'***, Click on '***Add-ins',*** Click on '***Solver Add-in'*** on the dialogue box). '***Solve***r' icon should appear on the right hand side of the menu bar when you click on '***Data'***).
7. Open the Solver window by clicking on '***Solver'***
8. Input the objective function using the 'cell reference' icon and select either 'Maximization' or 'Minimization'
9. Input the dummy values in Step 3 above as the 'changing variables'
10. Input the constraint equations using the appropriate icons as well as the resource limits
11. Click on 'solve'
12. Request for 'result' and 'sensitivity analysis'.

Example: Solution to LP Problem in Example 1









Steps Involved in Using Matrix Pivot Tool

1. Convert the constraint equations to thestandard form
2. Write the slack variables in the matrix format.
3. Add the other coefficients to the matrix

4. Place the cursor on pivot elements, i.e., the Basic Variables one by one and click '***Pivot on Selection***' the until you obtain your solution.

Example

Max P = 135 x + 90y

s.t.

$2x + y \leq 40$

$3x + 8y \leq 240$

$4x + 5y \leq 200$

$x,y \geq 0$

Solution

Convert the constraint equations to the standard form:

$2x + y + z_1 + 0z_2 + 0Z_3 = 40$

$3x + 8y + 0z_1 + z_2 + 0Z_3 = 240$

$4x + 5y + 0z_1 + 0z_2 + Z_3 = 200$

$x,y \geq 0$

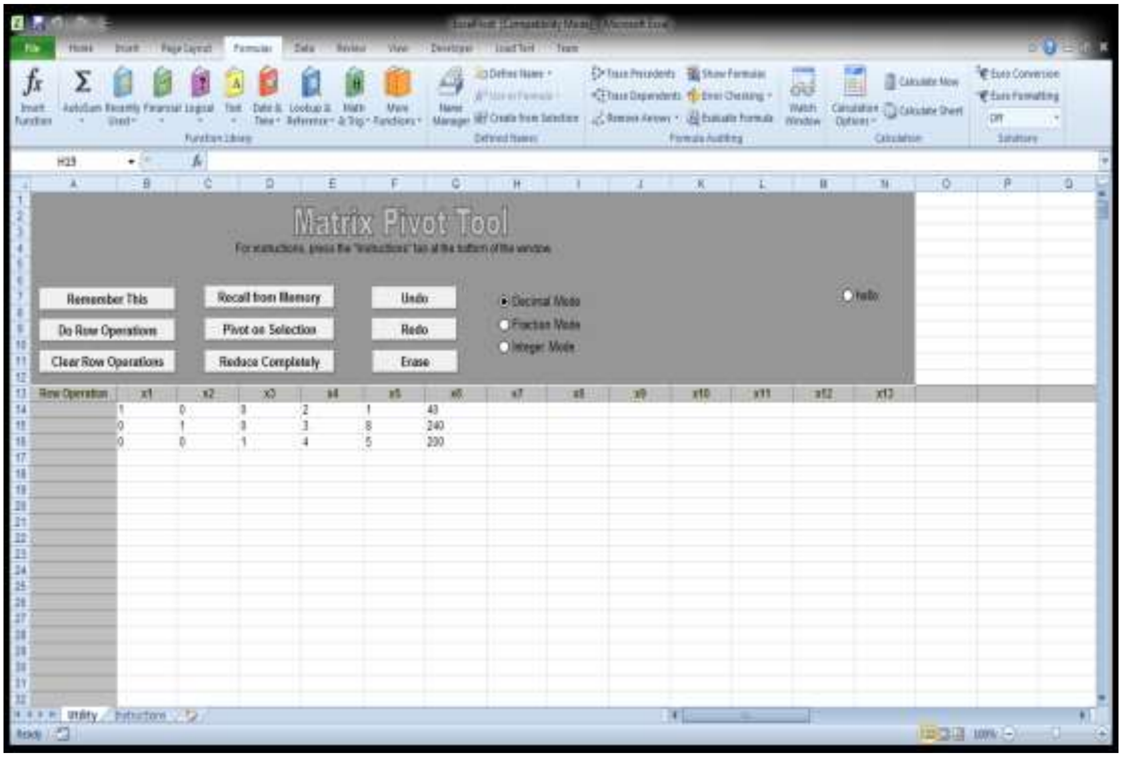
Convert the slack and constraint variables to a matrix

1 0 0 2 1 40

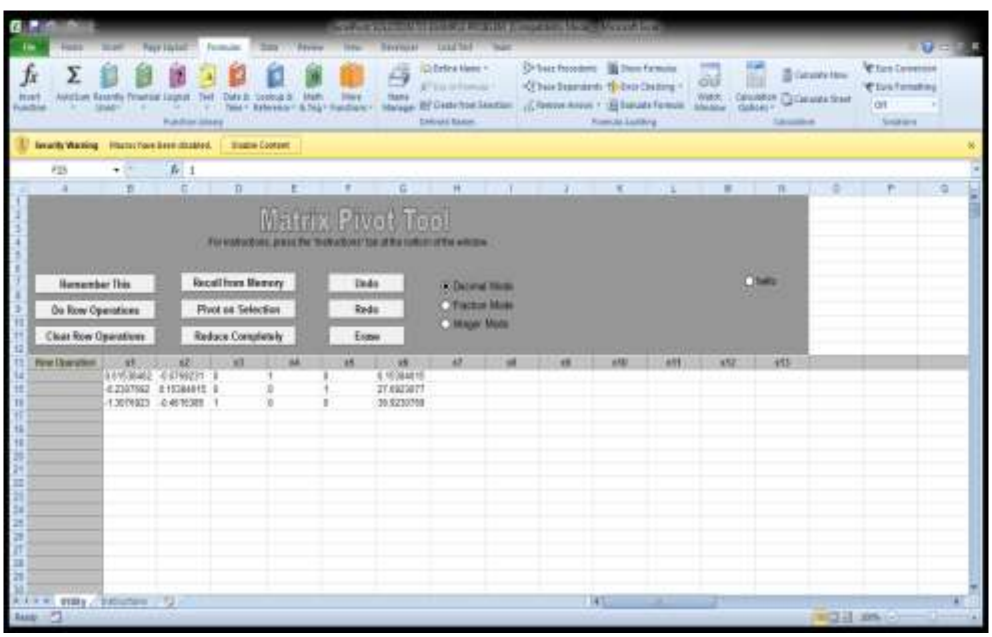
0 1 0 3 8 240

0 0 1 4 5 200

Tranfer the matrix to Excel Pivot Tool as shown below



Place the cursor on pivot elements, i.e., the Basic Variables one by one (which in this cases are the value of x in Row 1 and the value of y in Row 2 of the matrix) and click '***Pivot on Selection***' the until you obtain your solution as shown below:



As shown in the Matrix Pivot Tool, the solution variables are x = 6.15; y = 27.7
Hence, the Objective Function, P = 3323.25

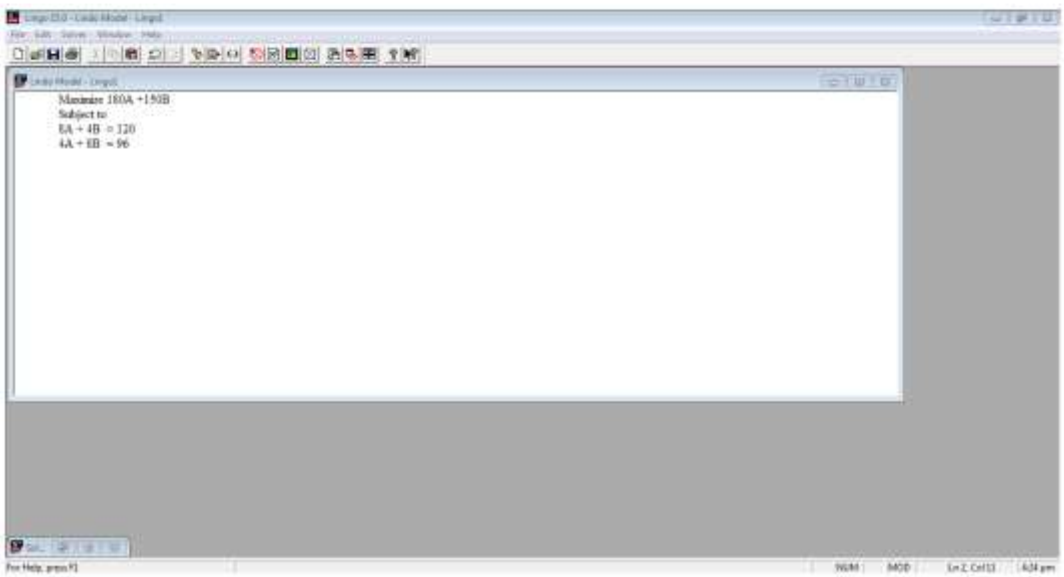
Steps Involved in Using *LINGO*

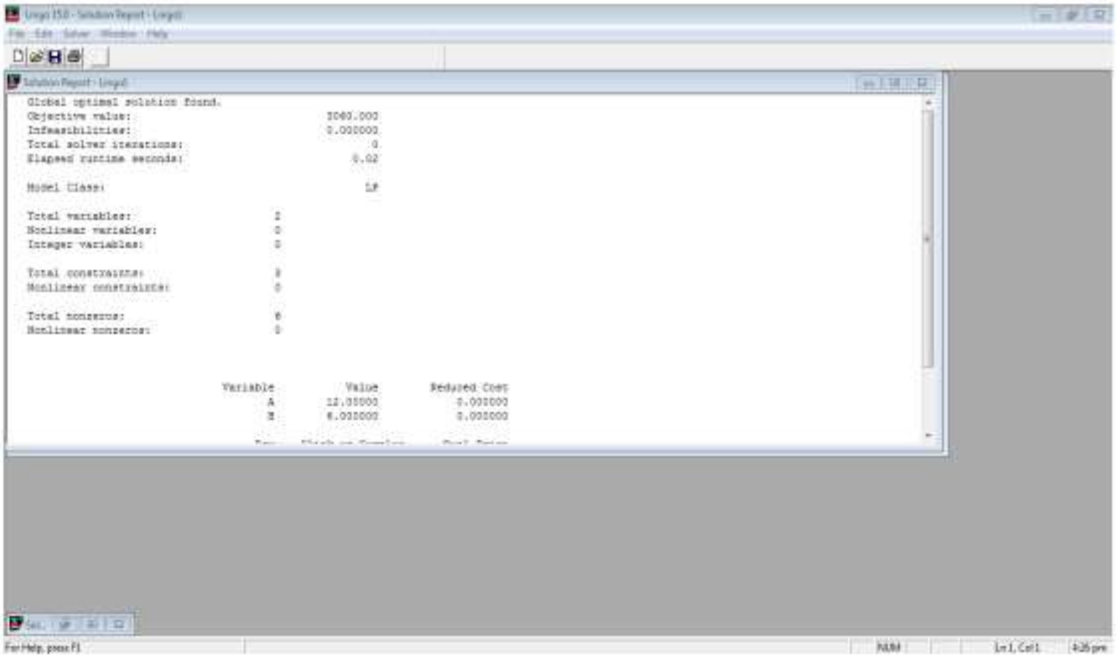
1. Open the *LINGO* window
2. Type in the objective function, i.e., Max $ax_1 + bx_2$ or Min $ax_1 + bx_2$
3. Type in the constraint equations, i.e.,
   Subject to

$$A_1x_1 + b_1x_2 \leq C_1$$
$$A_2x_1 + b_2x_2 \leq C_2$$

4. Click on 'Solve'

Example: Solution to LP Problem in Example 1





Exercises

1. A factory produces plywood and particleboard. The time to produce a unit of each product, the monthly capacity of operations and the net profit per unit of each product are as given in the table below.

i. Formulate the LP problem.

ii. Determine the optimum monthly production that will maximize profit using the graphical method.

iii. Construct the DUAL from the PRIMA .

iv. Use *EXCEL Solver* and *LINGO* software to solve LP problems.

Product Manufacturing Information

Operation	**Time per unit (hours)**		**Operation Capacity (hrs/month)**
	Plywood	**Particleboard**	
1	1	2	80
2	8	3	480
3	5	4	400
Net Profit per Unit (₦)	1350	1500	

2. A factory produces two products A and B. The time in hours to produce a unit of each product, the monthly capacity of operations in hours and the net profit per unit of each product in Naira are as given in Table 1 below.

v. Formulate the LP problem.

vi. Determine the optimum monthly production for the two products that will maximize profit using the graphical method.

vii. Construct the DUAL from the PRIMA and solve the DUAL using the Simplex method and *LINGO***.**

Product Manufacturing Information

viii.

Operation	**Time per unit (hours)**		**Operation Capacity (hrs/month)**
	Product A	**Product B**	
1	1	2	80
2	8	4	480
3	5	4	400
Net Profit per Unit (₦)	500	350	

ECO 413: OPERATIONS RESEARCH
LECTURE 3

OPTIMIZATION MODELS

2. TRANSPORTATION PROBLEM

Transportation problem is a special LP problem concerned with minimizing the cost of moving a given quantity of an item from specified sources to specifies locations, i.e., the movement of specified quantity of an item from i sources to j locations at minimum cost. There are several manual methods of solving the Transportation Problem, e.g., the North-West Corner Method, the Least Cost Method and Vogel's approximation Method. We shall demonstrate the uses of North-West Corner Method and Least Cost Method, though our focus in this course is the transformation of the problem to an LP format for computer solution.

North-West Corner Method

In this method, cell-to-cell allocation of items starts from the left hand corner, i.e. north-west corner of the cost allocation table, completely disregarding the unit transportation costs. The maximum number of items, based on the supply and demand is allocated from cell to cell until all supplies are exhausted and all demands are satisfied. The total cost is obtained by summing up the products of the number of the items allocated to cells and their respective unit costs.

Example

The table below shows the supply of plywood by a company from the Abuja, Sokoto and Kano depots of the company and the demand for plywood in the retail stores of the company in Lagos, Ibadan and Enugu. The figures in the cells are the costs in Naira of transporting one cubic metre of plywood from source *i* to location *j*. Determine the minimum transportation cost using the North-West Corner Method.

Transportation Cost Matrix

Source/Location	**Lagos**	**Ibadan**	**Enugu**	**Supply**
Abuja	800 X_{11}	700 X_{12}	600 X_{13}	**400**
Sokoto	1600 X_{21}	1000 X_{22}	900 X_{23}	**1200**
Kano	1900 X_{31}	1800 X_{32}	1200 X_{33}	**900**
Demand	**1400**	**950**	**150**	**2500**

Note

i. *The figures in the cells (800,700…1200) represent transportation costs per cubic metre of plywood*
ii. *The variables* X_{11}, X_{12}….. X_{33} *represent quantities of plywood*
iii. *The table shows that there are 400, 1200 and 900 cubic metres of plywood available in Abuja, Sokoto and Kano respectively, while the demands in Lagos, Ibadan and Enugu are 1400, 950 and 150 cubic metres respectively*

iv. *In this particular example, Total Supply = Total Demand = 2500 cubic metres of plywood.* ***This is referred to as the balanced case.***

Solution

Allocation Matrix (X_{ij}) Based on North-West Corner Method

Source/Location	**Lagos**	**Ibadan**	**Enugu**	**Supply**
Abuja	400			**400**
Sokoto	1000	200		**1200**
Kano		750	150	**900**
Demand	**1400**	**950**	**150**	**2500**

The associated transportation cost is given as:

P= ∑ cell alloction x unit cost

$P = \sum C_{ij} \times X_{ij}$

= (400 x 800) + (1000 x 1600) + (200 x 1000) + (750 x 1800) + (150 x 1200) = **₦3,650,000**

The Least Cost Method

In this method, the cell with the least unit cost is first considered for allocation. Allocation is done sequentially based on unit cost of transportation.

Example

Using the data in example given above for North-West Corner Method, the allocation will be as shown in the Table below:

Allocation Matrix (X_{ij}) Based on North-West Corner Method

Source/Location	**Lagos**	**Ibadan**	**Enugu**	**Supply**
Abuja		250	150	**400**
Sokoto	500	700		**1200**
Kano	900			**900**
Demand	**1400**	**950**	**150**	**2500**

The associated transportation cost is given as:

P= ∑ cell alloction x unit cost

$P = \sum C_{ij} \times X_{ij}$

= (250 x 700) + (150 x 600) + (500 x 1600) + (700 x 1000) + (900 x 1900) = **₦3,475,000**

Formulating Transportation Problem as a LP Minimization Problem

Let X_{ij} be the specified quantity of an item that should be transported from source i to location j at minimum cost

Let C_{ij} be the unit costs of transporting an item from source i to location j

The Constraints are the Demand and Supply Limits

Hence, the Objective Function is:

$\mathbf{P = \sum C_{ij} \times X_{ij}}$

Subject to:

$\mathbf{\sum X_{ij} \leq A}$ **(Demand Constraint)**

$\mathbf{\sum X_{ij} \leq B}$ **(Supply Constraint)**

Example

Using the data in example given for North-West Corner Method, the LP Formulation

Minimize $C=800x_{11} +700x_{12} + 600x_{13} + 1600x_{21} +1000x_{22} + 900x_{23} + 1900x_{31} + 1800x_{32} +1200x_{33}$

Subject to:

$x_{11} + X_{21}+ x_{31} \geq 1400$
$x_{12} + X_{22}+ x_{32} \geq 950$
$x_{13} + X_{23}+ x_{33} \geq 150$
$x_{11} + X_{12}+ x_{13} \geq 400$
$x_{21} + X_{22}+ x_{23} \geq 1200$
$x_{31} + X_{32}+ x_{33} \geq 900$
$x_{11}, X_{12} \ldots\ldots X_{33} \geq 0$

Solve the problem using LINGO and Excel Solver

Another Example

Formulate the transportation problem in the Table below as a Linear Programming minimization problem, i.e., i.e., state the equations for the objective function and constraints.

Transportation Cost Matrix

Source	**Location**				
	1	**2**	**3**	**4**	**Supply**
A	25	35	20	40	**250**
B	45	23	38	28	**400**
C	36	46	44	34	**100**
D	48	32	40	26	**550**
Demand	**650**	**50**	**350**	**250**	**1300**

Solution Using LP Formulation

The Objective Function is:

Minimize $C=25x_1 +35x_2 + 20x_3 + 40x_4 +45x_5 + \text{-------}+26x_{16}$

Subject to:

$x_1 + x_2 + x_3 + x_4 \geq 250$
$x_5 + x_6 + x_7 + x_8 \geq 400$
$x_9 + x_{10} + x_{11} + x_{12} \geq 100$
$x_{13} + x_{14} + x_{15} + x_{16} \geq 550$
$x_1 + x_5 + x_9 + x_{13} \geq 650$
$x_2 + x_6 + x_{10} + x_{14} \geq 50$
$x_3 + x_7 + x_{11} + x_{15} \geq 350$
$x_4 + x_8 + x_{12} + x_{16} \geq 250$
$x_1, x_2, \text{----------}x_{16} \geq 0$

Use LINGO and Excel Solver to solve the problem.

Unbalanced Case

This is a transportation problem in which the number of items supplied from all sources is not equal to the number of items demanded in all locations. When this situeation arises, a dummy source or location is created to balance the number. However, the cells in the dummy source or location is allocated zero unit transportation cost.With this amendment,the problem is solved as done in the Balanced case.

Maximization in a Transportation Problem

In certain cases of transportation problems the objective functin is to be maximised. This can be done by converting maximisation problem into minimisation problem.

Example

The Table below indicates the number of lorry loads of goods to be transported from sources 1,2,3,4 ,5 to locations A,B,C, and D. The figures in the cells of the table represent the unit costs in thousands of Naira of transporting a lorry load of goods from source i to location j. Formulate this problem as a LP Problem. Determine the total cost of transportation.

Cost of Transportation of Lorry Loads of Goods

	A	**B**	**C**	**D**	**Supply**
1	75	60	80	55	**250**
2	45	70	65	70	**850**
3	60	50	85	75	**750**
4	90	70	60	85	**350**
5	55	65	95	80	**300**
Demand	**150**	**450**	**900**	**500**	

Solution

Create a dummy location 'E' to balance the number of sources and locations; and allocate zero transportation unit cost to the cells in the dummy location as shown in the Table below.

	A	**B**	**C**	**D**	**E**	**Supply**
1	75	60	80	55	**0**	**250**
2	45	70	65	70	**0**	**850**
3	60	50	85	75	**0**	**750**
4	90	70	60	85	**0**	**350**
5	55	65	95	80	**0**	**300**
Demand	**150**	**450**	**900**	**500**	**500**	**2500**

Solve the problem Using (i) NorthWest Corner Method, Least Cost method, Excel Solver, and LINGO.

Transportation Problems with Prohibited Routes

Sometimes due to reasons like bad weather conditions, riots, strike, etc., some routes may not be available in a transportation problem. Then it is referred to as a problem with prohibited route(s). To handle such a problem, a very high cost is assigned to each of such routes which are not available. The problem is then solved in the usual way. Such high cost routes would automatically be eliminated in the final solution of the problem.

Exercises

1. A company wishes to transport its products from four sources to four locations. The unit costs of transportation (in '000 of Naira) are indicated in the Table below. Formulate the transportation problem as a Linear Programming minimization problem and solve the problem using LINGO and Excel Solver.

Transportation Cost Data

Source	**Location**				
	1	**2**	**3**	**4**	**Supply**
A	25	35	20	40	250
B	45	23	38	28	400
C	36	46	44	34	100
D	48	32	40	26	550
Demand	650	50	350	250	

1. The table below indicates the number of lorry loads of sand to be transported from sources 1,2,3,4 ,5 to locations A,B,C, and D. The figures in the cells of the table represent the unit costs in thousands of Naira of transporting a lorry load of particleboard from source i to location j. Determine the total cost of transportation using the North-West Corner, Minimum Cost methods and LINGO.

Cost of Transportation of Lorry Loads of sand

	A	**B**	**C**	**D**	**Supply**
1	75	60	80	55	**250**
2	45	70	65	70	**850**
3	60	50	85	75	**750**
4	90	70	60	85	**350**
5	55	65	95	80	**300**
Demand	**150**	**450**	**900**	**500**	

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LECTURE 3

OPTIMIZATION MODELS

3. ASSIGNMENT PROBLEM

When the problems involve the allocation of *n* different facilities to *n* different tasks, they are often termed assignment problems. The assignment problem can be better stated as follows: Given *n* facilities and *n* jobs, the problem is to assign each facility to one and only one job in a way that the measure of effectiveness is optimised.

Solution Steps

- Develop the cost Table 1 from the given problem. If the number of rows is not equal to the number of columns, a dummy row or column must be added to have a square matrix. The assignment cost of for dummy rows and columns are always zero.
- Find the **smallest cost (for minimization problems) or the largest cost (for maximization problems)** in each column of the cost Table 1.
- Subtract this cost element from each element in the column (ignoring the negative signs in the figures in the case of Maximization Problems) to produce Table 2.
- Find the smallest cost in each row of the REDUCED cost Table 2.
- Subtract this cost element from each element in that row to produce Table 3.
- Cross all the zeros in the cells of Table 3 with the MINIMUM NUMBER OF LINES. The lines may be vertical or horizontal. The lines may also intersect.
- If the number of lines in Step 6 = number of ruled lines, an optimal solution is reached. Go to Step 10. If not, take the following steps on Table 3.
- Examine those figures that are not covered by a line. Choose the smallest of these figures. Subtract this smallest figure from all the figures that that do not have a line through them. Add this smallest figure to every figure that lies at the intersection of two lines.
- Try to make the assignment again by crossing all zeros with the minimum number of horizontal and/or vertical lines. If the number of lines = number of ruled lines, an optimal solution is reached. If not repeat steps 8-9 as necessary.

10. Assign item to a zero which is unique to a row and/or a column.

Example1: Maximization Problem

Assuming the figures indicated in Table 4 (in '000 of Naira) are the service charges, assign jobs to the four engineers in a way that maximizes returns and determine the maximum return.

Assignment Problem Data

	Service Centre			
Location	**P**	**Q**	**R**	**S**
A	25	19	24	17
B	15	23	15	22
C	15	20	18	26
D	20	25	16	22

SOLUTION

Table 1- Subtract the largest figure in each column from every figure in the column, ignoring negative signs

	Service Centre			
Location	**P**	**Q**	**R**	**S**
A	0	6	0	9
B	10	2	9	4
C	10	5	6	0
D	5	0	8	4

- Tableau 2- Subtract the smallest figure in each row from every figure in the row.
- Cross all zeros with minimum number of lines. If the number of lines = number of assignments, carry out the assignment.

	Service Centre			
Location	**P**	**Q**	**R**	**S**
A	0	6	0	9
B	8	0	7	2
C	10	5	6	0
D		0	8	4

5

- If the number of lines is not equal to the number of assignments, identify the smallest uncrossed figure; subtract this figure from every uncrossed figure, add it to every figure crossed by 2 lines, leave figures crossed by a single line as they are. Cross all zeros with minimum number of horizontal/vertical lines

:

	Service Centre			
Location	**P**	**Q**	**R**	**S**
A	0	11	0	14
B	3	0	2	2
C	5	5	1	0
D	0	0	3	4

The Assignment that maximizes cost is as follows:

A $\longrightarrow$ R

B $\longrightarrow$ Q

C $\longrightarrow$ S

D $\longrightarrow$ P

Maximum return= ₦24000 + ₦23000 + ₦26000 + ₦20000 = ₦93000

Example 2: Minimization

Assuming the figures indicated in the original Table above (in '000 of Naira) are the costs of maintaining each engineer in each location assign jobs to the four engineers in a way that minimizes costs and determine the minimum cost.

SOLUTION

- Tableau 1- Subtract the smallest figure in each column from every figure in the column:

	Service Centre			
Location	**P**	**Q**	**R**	**S**
A	10	0	9	0
B	0	4	0	5
C	0	1	3	9
D	5	6	1	5

- Tableau 2- Subtract the smallest figure in each row from every figure in the row.
- Cross all zeros with minimum number of lines. If the number of lines = number of assignments, carry out the assignment. If the number of lines is not equal to the number of assignments, identify the smallest uncrossed figure; subtract this figure from every uncrossed figure, add it to every figure crossed by 2 lines, leave figures crossed by a single line as they are. Cross all zeros with minimum number of horizontal/vertical lines

	Service Centre			
Location	**P**	**Q**	**R**	**S**
A	10	0	9	0
B	0	4	0	5
C	0	$\boxed{1}$	3	9
D	4	5	0	4

	Service Centre			
Location	**P**	**Q**	**R**	**S**
A	11	0	10	0
B	0	3	0	4
C	0	0	3	8
D	4	4	0	3

The Assignment that minimizes cost is as follows:

A ⟶ S
B ⟶ P
C ⟶ Q
D ⟶ R

Minimum cost = ₦17000 + ₦15000 + ₦20000 + ₦16000 = ₦68000

Unbalanced Cases

This is an assignment problem in which the number of tasks is not equal to the number of facilities. When this situation arises, a task or facilitity is created to balance the number.

However, the cells in the dummy task or facilitity is allocated zero unit cost.With this amendment,the problem is solved as done in the Balanced case.

Prohibited Assignments

Sometimes technical, legal or other restrictions prohibit the assignment of any paricular facility to any particular job.. Then it is referred to as prohibited assignment. To handle such a problem, a very high cost is assigned to each of such assignments. The problem is then solved in the usual way. Such high cost assignments would automatically be eliminated in the final solution of the problem. In caseof a maximsation problem, the prohibited assignment are allocated very low profit.

Illustration

A wood workshop currently has five jobs to be done and has five machines to do them. The cost matrix gives thecost of processing each job on any machine. Because of specific job requirement and machine specifications, certain jobs cannot be done on certain machines. These have been shown by X in the cost matrix. The assignnent of jobs to machines must be done on a one-to-one basis. Assign the jobs to the machies in a manner that minimises the total cost wihin the stated restrictions

Cost Matrix

Machines	**Jobs**				
	1	**2**	**3**	**4**	**5**
A	80	40	X	70	40
B	X	80	60	40	40
C	70	X	60	80	70
D	70	80	30	50	X
E	40	40	50	X	80

Because certain jobs cannot be done on certain machines, we assign a high cost (say ₦500) to these cells and modify the cost matrix The modified cost matrix is given in the table below.

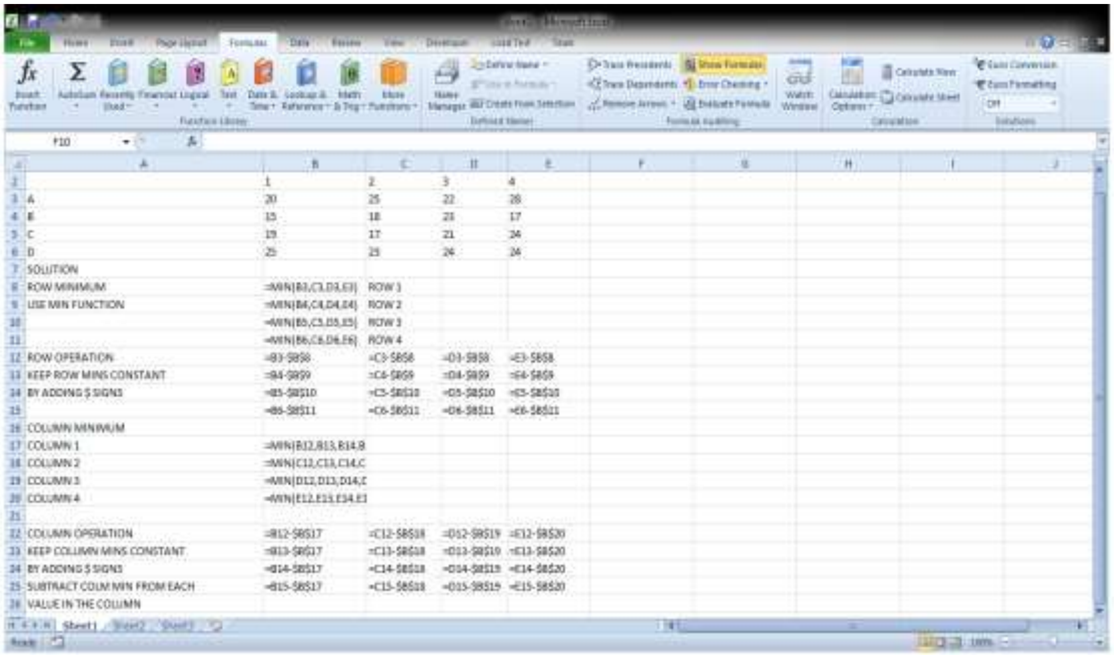
Modified Cost Matrix

Machines	**Jobs**				
	1	**2**	**3**	**4**	**5**
A	80	40	**500**	70	40
B	**500**	80	60	40	40
C	70	**500**	60	80	70
D	70	80	30	50	**500**
E	40	40	50	**500**	80

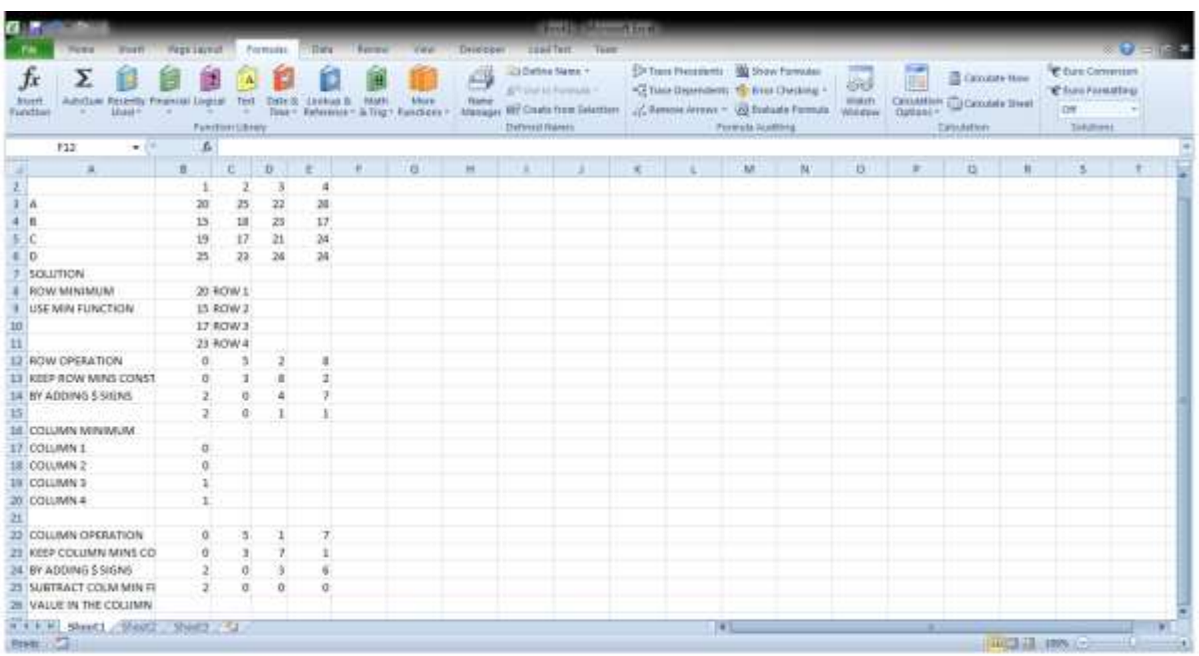
The problem can now be solved using the usual method.

An Excel Template for Solving Assignment Problems

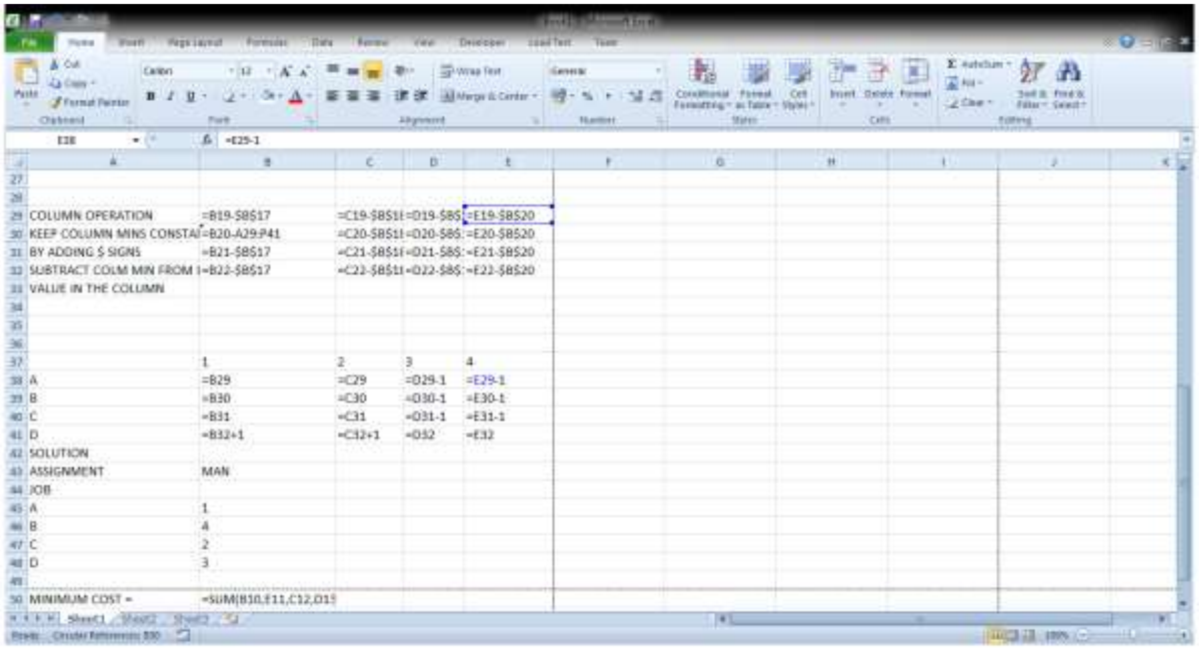
A job shop has 4 men available on 4 separate jobs. Only one man can work on any job. The cost of assigning each man to each job is given in the Table. The objective is to assign men to jobs such that the total cost of assignment is a minimum.

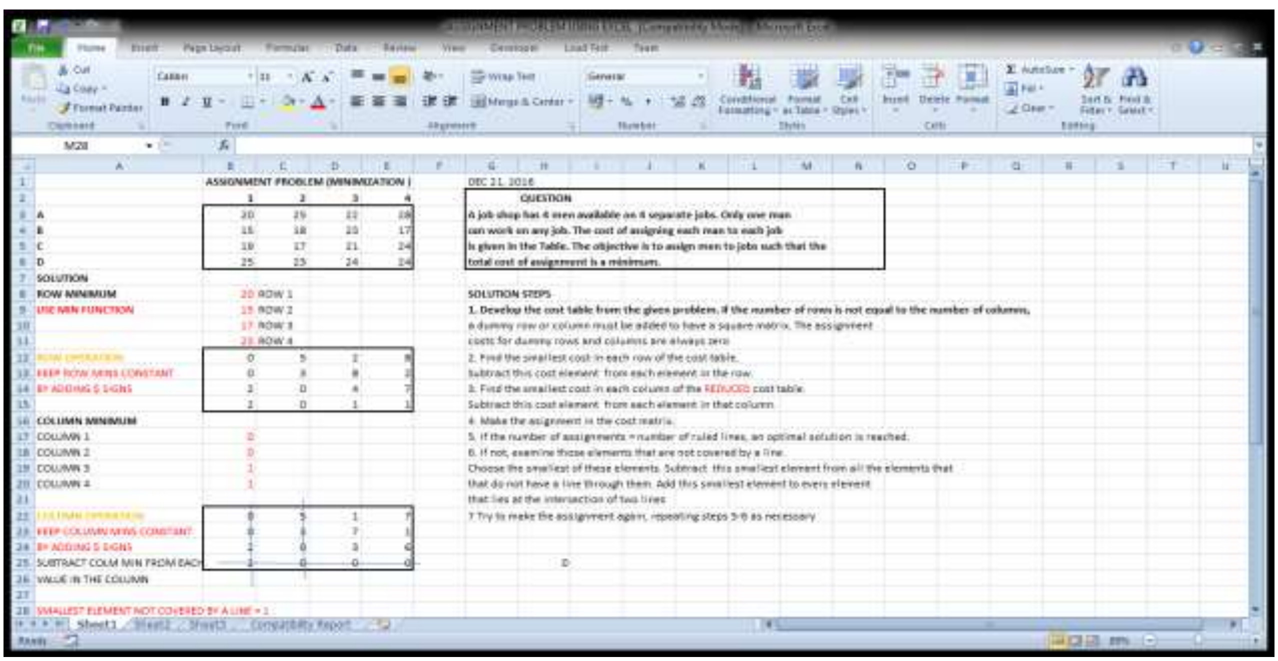


	A	B	C	D	E
2		1	2	3	4
3	A	20	25	22	28
4	B	15	18	23	17
5	C	19	17	21	24
6	D	25	23	24	24
7	SOLUTION				
8	ROW MINIMUM	=MIN(B3,C3,D3,E3)	ROW 1		
9	USE MIN FUNCTION	=MIN(B4,C4,D4,E4)	ROW 2		
10		=MIN(B5,C5,D5,E5)	ROW 3		
11		=MIN(B6,C6,D6,E6)	ROW 4		
12	ROW OPERATION	=B3-B8	=C3-B8	=D3-B8	=E3-B8
13	KEEP ROW MINS CONSTANT	=B4-B9	=C4-B9	=D4-B9	=E4-B9
14	BY ADDING $ SIGNS	=B5-B10	=C5-B10	=D5-B10	=E5-B10
15		=B6-B11	=C6-B11	=D6-B11	=E6-B11
16	COLUMN MINIMUM				
17	COLUMN 1	=MIN(B12,B13,B14,B			
18	COLUMN 2	=MIN(C12,C13,C14,C			
19	COLUMN 3	=MIN(D12,D13,D14,D			
20	COLUMN 4	=MIN(E12,E13,E14,E1			
21					
22	COLUMN OPERATION	=B12-B17	=C12-B18	=D12-B19	=E12-B20
23	KEEP COLUMN MINS CONSTANT	=B13-B17	=C13-B18	=D13-B19	=E13-B20
24	BY ADDING $ SIGNS	=B14-B17	=C14-B18	=D14-B19	=E14-B20
25	SUBTRACT COLUM MIN FROM EACH	=B15-B17	=C15-B18	=D15-B19	=E15-B20
26	VALUE IN THE COLUMN				



	A	B	C	D	E
2		1	2	3	4
3	A	20	25	22	28
4	B	15	18	23	17
5	C	19	17	21	24
6	D	25	23	24	24
7	SOLUTION				
8	ROW MINIMUM	20	ROW 1		
9	USE MIN FUNCTION	15	ROW 2		
10		17	ROW 3		
11		23	ROW 4		
12	ROW OPERATION	0	5	2	8
13	KEEP ROW MINS CONST	0	3	8	2
14	BY ADDING $ SIGNS	2	0	4	7
15		2	0	1	1
16	COLUMN MINIMUM				
17	COLUMN 1	0			
18	COLUMN 2	0			
19	COLUMN 3	1			
20	COLUMN 4	1			
21					
22	COLUMN OPERATION	0	5	1	7
23	KEEP COLUMN MINS CO	0	3	7	1
24	BY ADDING $ SIGNS	2	0	3	6
25	SUBTRACT COLM MIN FI	2	0	0	0
26	VALUE IN THE COLUMN				





Exercises

1. Table below shows the returns in thousands of Naira made by four sales representatives assigned to four different locations. Assign representatives in a way that maximizes returns.

Assignment Cost Data

Sales Rep	Location			
	1	2	3	4
Charles	58	66	69	55
Jude	72	93	48	82
Dave	85	70	76	46
Tola	68	75	56	78

2. The Table below shows the profit in hundreds of Naira made from the sales of four wooden furniture products in four different locations in Ibadan. Assign the products to different locations in a way that will maximize profit for the furniture maker.

Table showing Profit Made from Sales of Furniture Products

Furniture Product	Location			
	Agbowo	Bodija	Dugbe	Apata
Bed	78	68	56	75
Chair	82	72	48	93
Book Shelve	55	58	69	66
Shoe rack	46	85	76	70

3. A plywood manufacturing company has five jobs to be done but has four machines to do them. The cost of doing each job in thousands of Naira on each machine is presented in Table 3. Determine the job assignment which will minimize the total cost of and compute the minimum total cost.

Cost Information on Machine Jobs

Machine	Jobs				
	A	B	C	D	E
1	5	7	6	3	8
2	12	8	7	9	15
3	10	4	9	6	12
4	6	14	12	16	9

4. A manufacturing company has five jobs to be done on four machines. Each job must be done on one and only one machine. The cost of each job on each machine in thousands of Naira is given in the Table below. Because of the specific job requirements and machine specifications, certain jobs cannot be done on certain machines. These have been shown by X in the cost matrix. Allocate the job assignments to minimize cost and compute the minimum cost of doing the five jobs.

Cost Information on Machine Jobs

Job	Machine				
	1	**2**	**3**	**4**	**5**
A	80	40	X	70	40
B	X	80	60	40	40
C	70	X	60	80	70
D	70	80	30	50	X
E	40	40	50	X	80

ECO 413: OPERATIONS RESEARCH
LECTURE 5
JOB SCHEDULING AND SEQUENCING

Job scheduling refers to the process of determining when certain jobs are to be processed on given facilities or when the facilities are to be allocated to a given set of jobs. In simple terms, scheduling about the determination of:

- The rate at which the work shall be performed in order of priority
- The proper times when work will be released to the factory/plant
- The sequence to be followed by the jobs

Job sequencing refers to the order in which jobs are to be processed through the system or the order of processing the jobs through the facility or some facilities. In other words, sequencing is concerned with the selection of an appropriate sequence or order of performing a series of jobs on a finite number of machines (or service facilities) in some well-defined technological order so as to optimise some efficiency measure which may either be cost or time. The well-known sequencing problem is to determine the sequence in which two or more jobs should be processed on one or more machines in order to optimise a specific measure of efficiency.

For a sequencing problem, there should be the following elements:

- A set of two or more jobs to be done and a set of one or more machines on which to do the jobs.
- There is processing order of each job on each machine, i.e., one should know which machines are required to do each job and in what order these machines are required for the purpose.
- There is a given processing time.
- There should be a measure of efficiency or measure of effectiveness or measure of performance. The most common measure is minimisation of the total elapsed time.

Some Important Terms

i. **Total Elapsed Time**: The time between starting the first job and completing the last job.
ii. **Idle Time on a Machine**: The time a machine remains idle (without work) during the total elapsed time. Such a time can be worked out in respect of each machine.
iii. **Number of Machines**- means the service facilities through which job must pass or be serviced before it is completed.
iv. **Processing time**: The time each job requires on each machine.
v. **Processing Order**: The order in which various machines are required for completing the job.
vi. **Measure of Efficiency**: The goal to be optimised. Also commonly referred to as measure of effectiveness or measure of performance.
vii. **Priority Rule**: A rule for selecting which job is started first on some machine or work centre.

Assumptions in Case of Simple Sequencing Problems

i. Only one job can be processed on a given machine at a time

ii. The transportation time, i.e., the time involved in moving a job from one machine to another is very negligible and is assumed to be equal to zero.
iii. Each job, once started on a machine, is to be continued till its completion on the machine
iv. Machines to be used are of different types.
v. All jobs are fully known and are ready for processing.
vi. Processing times are given and do not change.
vii. Processing order is well-defined and meets the technological necessity.

Sequencing problems become complicated if the above stated assumptions do not hold good. Optimal sequencing procedures have been developed for the following situations:

1. 'n' jobs and one machine
2. 'n' jobs and two machines
3. 'n' jobs and three machines

There are no optimal rules for jobshops with more than two jobs and two machines which, in real life, is a common situation.

Priority Rules for Scheduling 'N' Jobs On One Machine (n/1 Problems)

This class of problems is referred to as 'n job-one machine problem' or simply n/1. The theoretical difficulty of this type of problem increases as more machines are considered rather than by the number of jobs that must be processed. Therefore, the only restriction on 'n' is that it must be a specified, finite number. The different priority rules available for solving this problem are the following:

i. **First come, first served (FCFS)-** the first job on the queue is serviced first, without consideration for due dates. No job is regarded as more important than the other. No consideration is given to due dates.
ii. **Shortest Processing Time/Shortest Operation Time (SPT/SOT):** Under this rule, the job with the least processing time is scheduled first, followed by the job having the next shortest processing time. This rule minimises the average flow time, including both waiting and service times for a particular job.
iii. **First in the System, First Served (FISFS):** The job that has stayed longest among remaining jobs is processed first. The rule tends to compensate jobs that have, for one reason or the other stayed long in the system.
iv. **Expected Due Date|(EDD):** A job with the closest due date is given priority. EDD rule is efficient in helping the shop to minimize lateness or tardiness.
v. **Slack:** The rule compares the due date for a particular job with the required processing time and schedules that job with the least slack time or period first. Slack can be positive or negative. A negative slack indicates that the job cannot be completed before due date. Utmost priority must be given to jobs with negative slacks.
vi. **Last Come, First Served**: This rule is rarely used in scheduling customers' orders. However, in warehouses where goods are stacked on pallets, the first commodity to arrive is stacked at the bottom and the last is stacked on top and processed first.

vii. Maximum Profit First: The job that gives the maximum profit/contribution to profit is processed first followed by the one with the next highest contribution to profit.

Example 1

Using FCFS and SPT/SOT rules, determine the total mean flow times for the jobs shown in the Table below.

Job Processing Times

Job (in order of arrival)	Processing time (Days)
A	8
B	6
C	7
D	5

SOLUTION

FCFS SCHEDULE

Job (in order of arrival)	Processing time (Days)	Flow Time/job
A	8	0+8 = 8
B	6	8+6 = 14
C	7	14+7 = 21
D	5	21 + 5 = 26

Total flow time = 26 days
Mean flow time = 26/4 = 6.5 days

SPT/SOT SCHEDULE

Job	Processing time (Days)	Flow Time/job
D	5	0+5= 5
B	6	5+6 = 11
C	7	11+7 = 18
A	8	18 + 8 = 26

Total flow time = 26 days
Mean flow time = 26/4 = 6.5 days

Example 2

Assume a single processor shop through which 10 jobs are to be processed. The processing time (P_i) for each job and the due dates are given below. Using the *MS WORD* sorting facility, determine the effectiveness of FCFS, SPT and EDD rules based on the following criteria:

i. Number of late jobs; ii. Total lateness; iii. Average lateness
iv. Total flow time for all jobs; v. Average flow time for all jobs
vi. Maximum lateness; vii. Minimum lateness

Job$_i$	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Estimated Processing Time P_i	1	2	9	9	4	6	7	3	8	2
Job Due Date D_i	5	2	6	8	10	4	1	3	7	12

Solution

i. **Using FCFS Rule** Jobs processed on basis of arrival time, No need for sorting.

Job_i	**P_i**	**Completion Time C_i**	**D_i**	**Lateness $L_i = C_i - D_i > 0$**
1	1	1	5	0
2	2	3	2	1
3	9	12	6	6
4	9	21	8	13
5	4	25	10	15
6	6	31	4	27
7	7	38	1	37
8	3	41	3	38
9	8	49	7	42
10	2	51	12	39
Total flow time for all jobs		218 Days		

i. Number of late jobs = 9
ii. Total lateness = 218 days
iii. Average lateness = 21.8 days
iv. Total flow time for all jobs= 272 days
v. Average flow time for all jobs= 27.2 days
vi. Maximum lateness = 42 days
vii. Minimum lateness =1day

ii. Using SPT Rule: Jobs processed on basis of processing times: Highlight and select columns1 and 2. Click on sorting icon and sort using Column 2 in ascending order

Job_i	**P_i**	**Completion Time C_i**	**D_i**	**Lateness $L_i = C_i - D_i > 0$**
1	1	1	5	0
2	2	3	2	1
10	2	5	12	0
8	3	8	3	5
5	4	12	10	2
6	6	18	4	14
7	7	25	1	24
9	8	33	7	26
3	9	42	6	36
4	9	51	8	43
Total flow time for all jobs		198 Days		

i. Number of late jobs = 8
ii. Total lateness = 151 days
iii. Average lateness = 15.1 days
iv. Total flow time for all jobs= 198 days
v. Average flow time for all jobs= 19.8 days

vi. Maximum lateness = 43 days
vii. Minimum lateness =1day

iii. Using EDD Rule: Jobs processed on basis of their due dates: Highlight and select columns1,2,3 and 4. Click on sorting icon and sort using Column 4 in ascending order

Job_i	**P_i**	**Completion Time C_i**	**D_i**	**Lateness $L_i = C_i - D_i > 0$**
7	7	7	1	6
2	2	9	2	7
8	3	12	3	9
6	6	18	4	14
1	1	19	5	14
3	9	28	6	22
9	8	36	7	29
4	9	45	8	37
5	4	49	10	39
10	2	51	12	39
Total flow time for all jobs		274 Days		

i. Number of late jobs = 10
ii. Total lateness = 216 days
iii. Average lateness = 21.6 days
iv. Total flow time for all jobs= 274 days
v. Average flow time for all jobs= 27.4 days
vi. Maximum lateness = 39 days
vii. Minimum lateness = 6 days

COMPARISON OF THE RESULTS OBTAINED WITH THE PRIORTY RULES

Priority Rule	**No of Late Jobs**	**Total Flow Time (days)**	**Total Lateness (days)**	**Average Flow Time (Days)**	**Maximum Lateness (days)**	**Minimum Lateness (days)**
FCFS	9	272	218	27.2	42	1
SPT	8	198	131	19.8	43	1
EDD	10	274	216	27.4	39	6

Practice Question

Assume a workshop through which 7 jobs are to be processed. The estimated processing time for each job, the due date and the unit contribution to profit are indicated in the table below. Determine the effectiveness of FCFS, SPT, EDD and Maximum contribution to profit rules using the following criteria: (a) No. of late jobs (b) Total lateness (c) Minimum total flow time (d) Maximum lateness.

Job_i	**P_i**	**D_i**	**Unit Contr. To Profit (N)**
1	9	8	500
2	4	10	650
3	6	4	700

4	7	1	450
5	3	3	800
6	8	7	250
7	2	12	550

Scheduling 'N' Jobs on Two Machines (n/2 Problems)

The next step up in complexity of jobshop types is the n/2 case where 2 or more jobs must be processed on 2 machines in a common sequence. As in the n/1 case, there is an approach that leads to an optimal solution according to certain criteria. This approach is termed ***Johnson's Rule*** (after its developer).

***Johnson's Rule* for Scheduling N Jobs on 2 Machines**

1. List all the jobs that are to be processed on the two machines, i.e., indicate the processing times of the n jobs.
2. Select the job which has the lowest processing time on both machines. If this lowest time is on Machine1, put it FIRST in the schedule. Otherwise (i.e., if it is on Machine2), put it LAST. Eliminate this job from further consideration. If there is a tie, select a job arbitrarily.
3. Apply Step 2 for all remaining jobs. The given sequence is optimal.
4. Calculate the total elapsed time and idle times of the jobs and machines, if any.

Example 1

Assume that the information relating to 4 jobs and the sequence requirements in a 2-machine jobshop are given in the table below. Jobs 1 – 4 must be processed first on Machine A and then Machine B. Determine the optimal schedule and the total elapsed time.

Job	**Processing Times on Machines**	
	Machine A	**Machine B**
1	3	3
2	2	4
3	1	3
4	2	6

Solution- using Johnson's Rule

i. **Optimal Schedule**

Job	**Scheduled Order**
3	1
2	2
4	3
1	4

ii. **Gantt Load Chart for Computing the Total Elapsed Time:** Gantt chart is a graphical display of the job sequencing. It is very useful in determining the total flow time (makespan), machine and job waiting times.

Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
Machine A	3	2	2	4	4	1	1	1									
Machine B		3	3	3	2	2	2	2	4	4	4	4	4	4	3	3	3

Total Elapsed Time = 17 Days

Example 2

Shown in the table below is a scheduling problem in a factory. All jobs must pass through the inspection unit before going to the packaging and shipping unit. Determine (a) the optimal schedule; (b) the total elapsed time; (c) total machine waiting for jobs; and (d) the total machine idle time.

Job	**Processing Times**	
	Inspection Unit	**Packaging & Shipping Unit**
A	1	3
B	6	4
C	5	7
D	2	2
E	6	6

Iteration 1:

A				

Iteration 2: (Job D was arbitrarily scheduled)

A				D

Iteration 3:

A			B	D

Iteration 4:

A	C		B	D

Iteration 5:

A	C	E	B	D

i. **Optimal Schedule**

Job	**Scheduled Order**

A	1
C	2
E	3
B	4
D	5

i. **Gantt Load Chart for Computing the Total Elapsed Time**

Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Inspection	A	C	C	C	C	C	E	E	E	E	E	E	B	B	B	B	B	B	D	D	-	-	-	-	-
Packaging	-	A	A	A	-	-	C	C	C	C	C	C	C	E	E	E	E	E	E	B	B	B	B	D	D

Total Elapsed Time = 25 Days

Summary of Waiting Idle Time

Job	Inspection	Packaging
A	-	-
C	-	-
E	-	1
B	-	1
D	-	3

Total waiting time = 5 units of time

Machine Waiting for Job

	A	B	C	D	E	Total
Inspection	-	-	-	-	-	0
Packaging	1	-	2	-	-	3

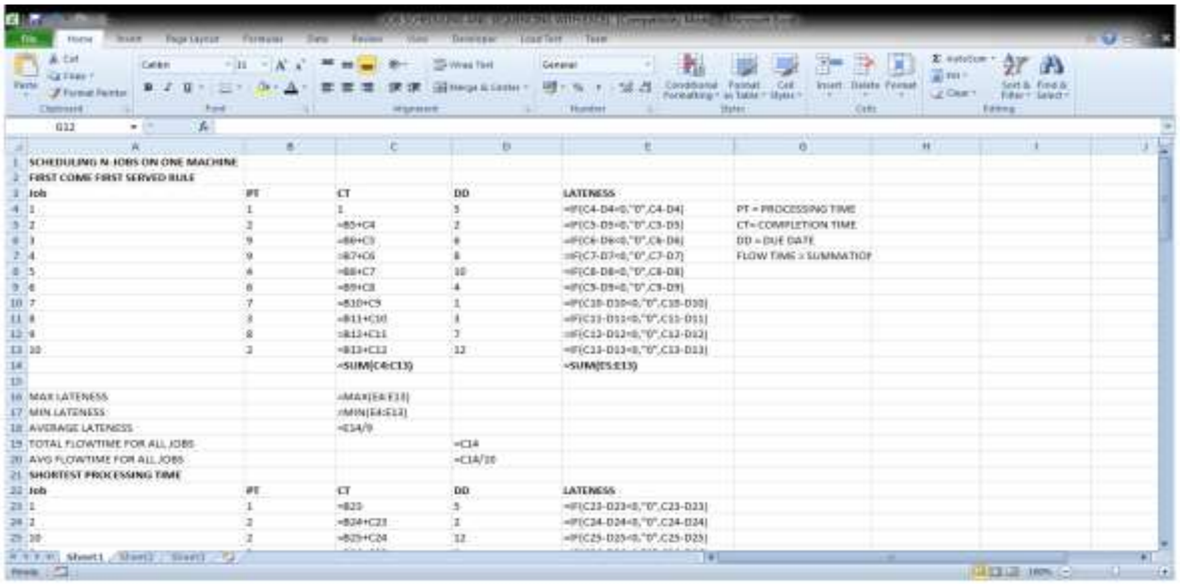
Total machine idle time = 3 units of time

Summary of Results

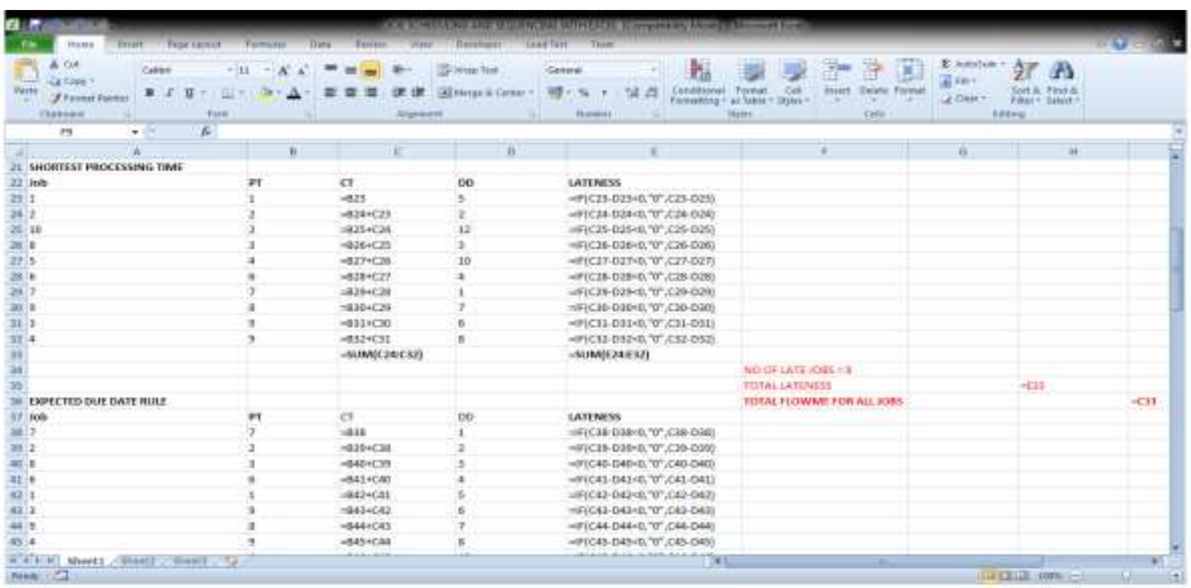
In all, the total makespan for the entire job is 25 units of time. There is no idle time on Machine 1 (i.e., inspection unit). Jobs B, D and E incurred a total of 5 units of waiting time. There are 3 units of idle time on Machine 2 (i.e., packaging and shipping unit).

Use of Excel for Job Scheduling and Sequencing

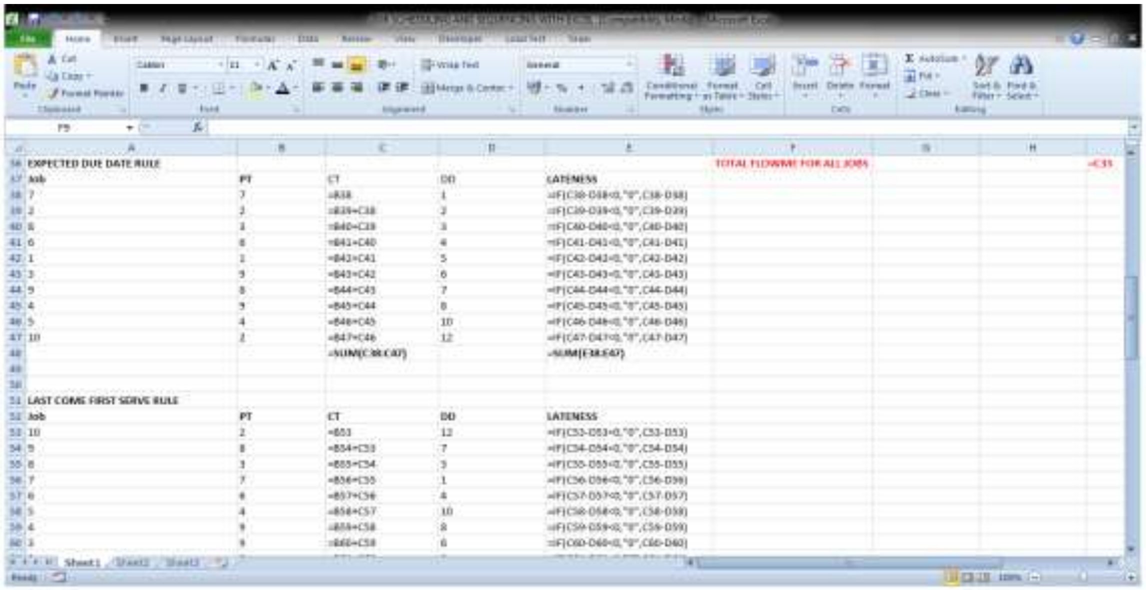
Apart from MS Word, MS Excel can also be used to solve job scheduling and sequencing problems as shown in the template below



	A	B	C	D	E	G
1	**SCHEDULING N-JOBS ON ONE MACHINE**					
2	**FIRST COME FIRST SERVED RULE**					
3	**Job**	**PT**	**CT**	**DD**	**LATENESS**	
4	1	1	1	5	=IF(C4-D4<0,"0",C4-D4)	PT = PROCESSING TIME
5	2	2	=B5+C4	2	=IF(C5-D5<0,"0",C5-D5)	CT= COMPLETION TIME
6	3	9	=B6+C5	6	=IF(C6-D6<0,"0",C6-D6)	DD = DUE DATE
7	4	9	=B7+C6	8	=IF(C7-D7<0,"0",C7-D7)	FLOW TIME = SUMMATIOF
8	5	4	=B8+C7	10	=IF(C8-D8<0,"0",C8-D8)	
9	6	6	=B9+C8	4	=IF(C9-D9<0,"0",C9-D9)	
10	7	7	=B10+C9	1	=IF(C10-D10<0,"0",C10-D10)	
11	8	3	=B11+C10	3	=IF(C11-D11<0,"0",C11-D11)	
12	9	8	=B12+C11	7	=IF(C12-D12<0,"0",C12-D12)	
13	10	2	=B13+C12	12	=IF(C13-D13<0,"0",C13-D13)	
14			**=SUM(C4:C13)**		**=SUM(E5:E13)**	
15						
16	MAX LATENESS		=MAX(E4:E13)			
17	MIN LATENESS		=MIN(E4:E13)			
18	AVERAGE LATENESS		=E14/9			
19	TOTAL FLOWTIME FOR ALL JOBS			=C14		
20	AVG FLOWTIME FOR ALL JOBS			=C14/10		
21	**SHORTEST PROCESSING TIME**					
22	**Job**	**PT**	**CT**	**DD**	**LATENESS**	
23	1	1	=B23	5	=IF(C23-D23<0,"0",C23-D23)	
24	2	2	=B24+C23	2	=IF(C24-D24<0,"0",C24-D24)	
25	10	2	=B25+C24	12	=IF(C25-D25<0,"0",C25-D25)	



	A	B	C	D	E	F	H	I
21	**SHORTEST PROCESSING TIME**							
22	**Job**	**PT**	**CT**	**DD**	**LATENESS**			
23	1	1	=B23	5	=IF(C23-D23<0,"0",C23-D23)			
24	2	2	=B24+C23	2	=IF(C24-D24<0,"0",C24-D24)			
25	10	2	=B25+C24	12	=IF(C25-D25<0,"0",C25-D25)			
26	8	3	=B26+C25	3	=IF(C26-D26<0,"0",C26-D26)			
27	5	4	=B27+C26	10	=IF(C27-D27<0,"0",C27-D27)			
28	6	6	=B28+C27	4	=IF(C28-D28<0,"0",C28-D28)			
29	7	7	=B29+C28	1	=IF(C29-D29<0,"0",C29-D29)			
30	9	8	=B30+C29	7	=IF(C30-D30<0,"0",C30-D30)			
31	3	9	=B31+C30	6	=IF(C31-D31<0,"0",C31-D31)			
32	4	9	=B32+C31	8	=IF(C32-D32<0,"0",C32-D32)			
33			**=SUM(C24:C32)**		**=SUM(E24:E32)**			
34						NO OF LATE JOBS = 8		
35						TOTAL LATENESS	=E33	
36	**EXPECTED DUE DATE RULE**					**TOTAL FLOWME FOR ALL JOBS**		=C33
37	**Job**	**PT**	**CT**	**DD**	**LATENESS**			
38	7	7	=B38	1	=IF(C38-D38<0,"0",C38-D38)			
39	2	2	=B39+C38	2	=IF(C39-D39<0,"0",C39-D39)			
40	8	3	=B40+C39	3	=IF(C40-D40<0,"0",C40-D40)			
41	6	6	=B41+C40	4	=IF(C41-D41<0,"0",C41-D41)			
42	1	1	=B42+C41	5	=IF(C42-D42<0,"0",C42-D42)			
43	3	9	=B43+C42	6	=IF(C43-D43<0,"0",C43-D43)			
44	9	8	=B44+C43	7	=IF(C44-D44<0,"0",C44-D44)			
45	4	9	=B45+C44	8	=IF(C45-D45<0,"0",C45-D45)			



Practice Questions

1. In a factory, there are 6 jobs to be performed, each of which should go through machines M1 and M2in the order M1, M2. The processing times (hours) for the jobs are given below. Determine the sequence for performing the jobs that would minimize the total elapsed time and find its value.

Job	1	2	3	4	5	6
M1	1	3	8	5	6	3
M2	5	6	3	2	2	10

2. Five jobs are to be processed on two machines A and B in an integrated mill in the order A,B. Processing times in hours are given below. Determine the sequence that will minimize the total elapsed time.

Job	1	2	3	4	5
Machine A	5	1	9	3	10
Machine B	2	6	7	8	4

3. A factory has to take 7 products through two manufacturing processes. The time required for the two processes are given below. Determine the sequence in which the products should be processed to minimize the total elapsed time.

Log	1	2	3	4	5	6	7
Process 1	5	7	3	4	6	7	12
Process 2	2	6	7	5	9	5	8

Scheduling 'N' Jobs on Three or More Machines

In situations where there are three or more machines through which the jobs must pass, Johnson's Rule can be used by collapsing the processing times on the 3 or 'n' machines into 2-machine processing times.

Example

Five jobs are to be assigned to three machines. The processing times in days on the three machines are as shown in the Table below. Using Johnson's rule, determine the optimal job schedule and the total elapsed time, job and machine waiting times (if any).

Job and Machine Processing Times (Days)

Job	Machines		
	M1	M2	M3
1	6	5	4
2	2	1	3
3	6	2	4
4	5	2	1
5	2	5	3

SOLUTION

i. Collapse the processing times on the 3 machines into 2-machine processing times as follows:

Job	M1 + M2	M2 +M3
1	11	9
2	3	4
3	8	6
4	7	3
5	7	8

ii. **Optimal Job Schedule:** Apply Johnson's Rule as follows- select the job that has the lowest processing time on both machines. If this lowest time is on machine 1, put it first in the schedule, otherwise (i.e., if it is on machine 2) put it last. Eliminate this job from further consideration. If there is a tie, select a job arbitrarily.

Job 2	Job 5	Job 1	Job 3	Job 4

iii. **The total elapsed time:** Using Gantt load chart (not shown here because of length), the total elapsed time = 39 days

iv. **Job Waiting time:** As indicated on the Gantt load chart, the total job waiting time = 1 day for Job 3

v. **Machine waiting:** As indicated on the Gantt load chart, the total machine waiting time = 9 days for machine 2

ECO 413: OPERATIONS RESEARCH
LECTURE 6
SIMPLE QUEUING/WAITING LINE PROBEMS

A queue is an array of items/customers waiting to be served, e.g., logs of wood in a sawmill, items or customers in a wood workshop, etc. Waiting for service is part of daily life. In other words, queues are very common in everyday life. Also, the waiting phenomenon is not an experience limited to human beings: Jobs wait to be processed on a machine, planes circle in stack before being given permission to land, and cars stop at traffic lights. Queuing models are those where a facility performs a service. A queue problem arises when the service rate of a facility falls short of the flow rate of customers. If the service facility is capable of servicing the customer when he arrives, no bottlenecks/waiting line will occur. But if it takes 15 minutes to service a customer and one customer arrives every 12 minutes, then a queue will build up to a finite length of the same arrival and service rates continue. In such a situation the bottleneck is eliminated only if arrival rate decreases or service rate increases or there is an increase in the number of service facilities.

Terms commonly used in Queuing Theory/Model

1. **Customer**: Persons or units arriving in a station for service. Customers may be either persons or machines or other items. Customers arrive at a service facility from a source. On arrival, a customer can start receiving service immediately or wait in a queue if the facility is busy. When a facility completes a service, it automatically ‘pulls’ a waiting customer, of any, from the queue. If the queue is empty, the facility becomes idle until a new customer arrives.
2. **Service Station/Service facility**: Point where service is provided.
3. **Waiting Time**: Time a customer spends in the queue before being serviced.
4. **Time spent by a Customer in the System**: Waiting time plus service time.
5. **Number of Customers in the System**: Number of customers in the queue plus the number of Customers being serviced.
6. **Queue Length**: Number of customers waiting in the queue.
7. **Queue discipline**: The order in which customers are selected from a queue. The most common discipline is First Come, First Served (FCFS). Other disciplines include Last Come, First Served (LCFS) and Service in Random Order (SIRO). Customers may also be selected from the queue based on some order of priority. For example, rush jobs in a shop are processed ahead of regular jobs.
8. **Jockeying**: Joining the other queue and leaving the first one.
9. **Reneging**: Joining the queue and leaving it afterwards.
10. **Balking**: Deciding not to join the queue.

Object of the Queuing Theory

If there are queues then customers have to wait for some time before service. However, eliminating waiting altogether is not a feasible option because the cost of installing and operating the service facility can be prohibitive. The time lost in waiting is often expensive in terms of money, equipment, etc. As such, there are costs associated with waiting in line, commonly known as ***waiting-time costs***. On the other hand, if there are no queues, members of the servicing station might stay idle. Costs associated with service or facility are known as ***service costs***. The only recourse is to strike a balance between cost of offering a service and the cost of waiting experienced by customers. Queuing analysis is the vehicle for achieving this goal. The object of the queuing theory, therefore, is to achieve a good economic balance between these two costs. The optimum solution is when the sum of the waiting line and service costs is the minimum, i.e., the object of the queuing theory is to minimise the total waiting and service costs.

Note: The queuing theory does not directly solve the problem of minimising the total waiting and service costs. It only provides information necessary to take relevant management decisions.

Queue Characteristics

Every queue has four component parts:

i. **Arrival Pattern**: It is important to know the number of customers expected for service and how they arrive. It is equally important to know the time intervals between arrivals (i.e., time between successive arrivals or inter-arrival time) and the average arrival per unit time. ***In this course we shall treat arrival patterns first as a certain entity and later as a probability during simulation***.

ii. **Queue** – waiting to be served. The type of queue and the guideline for queuing are important. It is important to know whether or not a customer can quit (reneging) a queue or cross from one queue to another (jockeying). It is also important to know whether or not there is a limit to the length of the queue.

iii. **Service Pattern** It is important to know the number of service points and the number of customers that can be served simultaneously. The number of customers that can be served per unit time and the type of services available are also important. There are four types of queue-service relationships:
   - **Single queue to single service point**, e.g. one type of log queuing for service on a bandmill.
   - **Single queue to many service points**, e.g. one type of log queuing for service on two or more bandmills.
   - **Many queues to single service point**, e.g. different types of logs queuing for service on a bandmill.
   - **Many queues to many service points** e.g. different types of logs queuing for service on two or more bandmills.

iv. **Departure**- The number of customers departing per unit time and the rules governing the departure are also important.

Simple Queue Formulas

i. **Traffic Intensity:** We define traffic intensity (ρ) as

$$\rho = \frac{\text{arrival per unit time}}{\text{service per unit time}} = \frac{\beta}{\mu}$$

Note: If the traffic intensity is greater than 1, then the queue will keep growing indefinitely without end. If the traffic intensity is equal to 1, then no change in the queue length will be noticed. If the traffic intensity is less than 1, then ***this is a simple queue*** whose length will diminish gradually. ***The focus of this course is on such simple queues*** to which all the following formulas are applicable.

ii **Average number in Queue (queue or no queue):** This is the number of items in queue whether or not there is a queue.

$$\tilde{N} = \frac{\rho^2}{1-\rho}$$

iii. **Average number in Queue (when queue exists):** This is the number of items in queue when there is a queue.

$$\tilde{N}_1 = \frac{1}{1-\rho}$$

iv. **Average number in system:** This is the number of items in queue and receiving service:

$$S = \frac{\rho}{1-\rho}$$

v. **Average time in queue (before service is rendered):**

$$w = \frac{\rho}{1-\rho} \times \frac{1}{\mu}$$

vi. Average time in the system (on queue and receiving service):

$$t = \frac{1}{1-\rho} \times \frac{1}{\mu}$$

vii. **Average time of service = t – w**

viii. **Probability of queuing on arrival**: This is equal to the traffic intensity, ρ.

ix. **Probability of not queuing on arrival:** This is equal to 1-ρ.

x. **The probability that there are n items in the system**: This is given as

$$\text{Prob } (x=n) = (1-\rho)\rho^n$$

xi. **The probability that there are more than n items in the system**: This is given as

$$\text{Prob } (x>n) = \rho^n$$

Worked Examples

1. In a factory, the arrival of customers per hour is 12 while the number served per hour is 18. Find the traffic intensity.

   Solution

   Arrival rate, β = 12
   Service rate, μ = 18
   Traffic intensity, P = 12/18 = 2/3

2. In a simple queue, the arrival is 20 in 4 hours while the service is 48 in 8 hours. Determine the traffic intensity.
   Solution
   Arrival rate, $\beta = 20/4 = 5$ per hour
   Service rate, $\mu = 48/8 = 6$ per hour
   Traffic intensity, $P = 15/6$

3. In a simple queue system, the arrival rate is 80 in 5 hours while the service is 120 in 10 hours. Determine the traffic intensity. Does the system agree with simple queue characteristics?
   Solution
   Arrival rate, $\beta = 80/5 = 16$ per hour
   Service rate, $\mu = 120/10 = 12$ per hour
   Traffic intensity, $P = 16/12 = 4/3$
   Since P > 1, the system does not agree with simple queue characteristics.

4. Given that the traffic intensity in a simple queue (queue or no queue) is 2/3, find the number of items in the queue.
   Solution
   $$\tilde{N} = \frac{\rho^2}{1-\rho}$$
   $= (2/3)^2 / (1-2/3) = 4/3 = 1.3$ (= 1 since N is discrete)

5. A company distributes its products to customers in a queue at an average rate of 25 per week. Demand for the company's products is estimated at an average of 20 per week. Compute (a) the traffic intensity and (b) the average number of customers in queue (queue or no queue).
   Solution
   (a) Traffic intensity, $\rho = 20/25 = 4/5$
   (b) Average No.in queue $\tilde{N} = \dfrac{\rho^2}{1-\rho}$
   $= (4/5)^2 / (1-4/5) = 16/5 = 3$ customers (Since N is discrete)

6. A company distributes its products to customers in a queue at an average rate of 25 per hour. Demand for the company's products is estimated at an average of 20 per hour. Compute (a) the traffic intensity, (b) the average number of customers in queue (when queue exists), and (c) average time in queue (before service is rendered).
   Solution
   (a) Traffic intensity, $\rho = 20/25 = 4/5$
   (b) Average No.in queue $\tilde{N}_1 = \dfrac{1}{1-\rho}$
   $= 1/(1-0.8) = 5$customers
   (c) Average time in queue (before service is rendered):

   $w = \underline{\rho} \quad x \quad \underline{1}$

1- ρ μ

(0.8)/ (1-0.8)] x (1/25) = 0.16 hours

ECO 413: OPERATIONS RESEARCH
LECTURE 7

SIMULATION

Simuation modelling is used to solve problems involving conditions of uncertainty and where mathematical formuation is impossible. For example, the assumption of ***constant arrival and service rates*** made in the queuing theory makes quite easy the calculation of whether a queue will form and what its length will be after any period of time. However, the more usual case is one in which ***both arrivals and services are not constant but are randomly distributed***. Randomness in simulation arises when the interval between successive events is probabilistic. One method of treating such problems is by Monte Carlo simulation.

Monte Carlo simulation is conducted using random numbers. A table of random numbers is a group of numbers which is not arranged in any order, i.e., they are all mixed up and no one number is more likely to occur next than any other number. Random numbers can be generated in different ways, including the folowing:

- Throwing coins or dice
- Using random number tables
- Using the random number generating facility of a calculator
- Using the random number generating facility of ***Excel*** to creat number tables
- Using pseudo-random number algorithms
- Writing a computer program to generate the random numbers

Example of Monte Carlo Simulation using Coins

A company operates on contract basis. It may or may not receive jobs, i.e. it has a 50-50 chance of receiving jobs on a daily basis. Simulate the situation for 20 days by assuming each day's business based on the toss of a coin for getting jobs, assume that *Head* means jobs are available and *Tail* means no job. Compare the theoretical probability with the simulated result.

Solution

Solution Steps

1. Toss a coin 20 times, a toss representing 1 day.
2. Each time, check and record the side of the coin that is displayed.
3. Compute the number of times each side is displayed.

The toss of a coin 20 times yielded the following result:

T,T,H,T,T,H,T,H,H,T,H,H,T,H,T,T,H,H,H,T

There are 10 Heads = 10 days of job availability (50%)

There are 10 Tails = 10 days of No job (50%)

Hence, the theoretical probability compares favourably with the simulated result

Example of Monte Carlo Simulation using the Dice

Assuming the same company above receives jobs for 10 days. However, the job may be small 1/3 times, medium 1/3 times and big 1/3 times. Simulate the situation for 10 days with the throw of a dice, assuming that getting *1 or 2* means small jobs are supplied, getting *3 or 4* means medium jobs are supplied and getting *5 or 6* means big jobs are supplied. Compare the theoretical probability with the simulated result.

Solution Steps

1. Throw a dice 10 times, a throw representing 1 day.
2. Each time, check and record the figure that is displayed.
2. Compute the number of times each pair of figures is displayed.

The throw of a dice 10 times yielded the following result:
2,3,5,6,4,3,5,1,6,1
1 or 2 = 3 times = 3 days of small jobs (0.3)
3 or 4 = 3 times = 3 days of medium jobs (0.3)
5 or 6 = 4 times = 4 days of big jobs (0.4)
Hence, the theoretical probability compares favourably with the simulated result

Example of Monte Carlo Simulation using a Calculator

Assume that the data presented in the table below were collected from the survey of the operations of a bank. Conduct a 20-day simulation of daily customer arrival using scientific calculator to generate random numbers and determine total number of customers.

Log Arrival Data

Customer arrival/day	Frequency of observation
16	8
20	6
25	7
40	10
10	12
36	4

Solution Steps

1. Re-arrange the customer arrival per day data and their associated frequencies in ascending order, from the smallest to the highest figure (as shown in Column 1 of Table A).
2. Compute the total frequency (as shown in the last row of Column 2 of Table A).
3. Compute the probabilities (as shown in Column 3 of Table A).
4. Compute the probability densities, i.e., cumulative frequency (as shown in Column 4 of Table A).
5. Set up the random number range for each cumulative frequency of each observation (as shown in Column 5 of the Table A).
6. Generate random number with your calculator, set up a random number table and conduct the simulation as shown in Table B

Table A

Customer arrival/day	Frequency of Observation	f(X)	Cumulative Frequency F(X)	Random Number Range
10	12	12/47	12/47 =0.26	0-0.26
16	8	8/47	20/47 = 0.43	0.27- 0.43
20	6	6/47	26/47 = 0.55	0.44-0.55
25	7	7/27	33/47 = 0.70	0.56 – 0.70
36	4	4/47	37/47 = 0.79	0.71 – 0.79
40	10	10/47	47/47 =1.00	0.80 – 1.00
Total Frequency	**47**			

Table B: Simulation Table

Day	Random Number	No. of customers
1	0.019	10
2	0.79	36
3	0.87	40
4	0.81	40
5	0.22	10
6	0.15	10
7	0.62	25
8	0.65	25
9	0.22	10
10	0.94	40
11	0.22	10
12	0.76	36

13	0.79	36
14	0.74	36
15	0.75	36
16	0.51	20
17	0.05	10
18	0.51	20
19	0.13	10
20	0.97	40
Total No of jobs		**500**

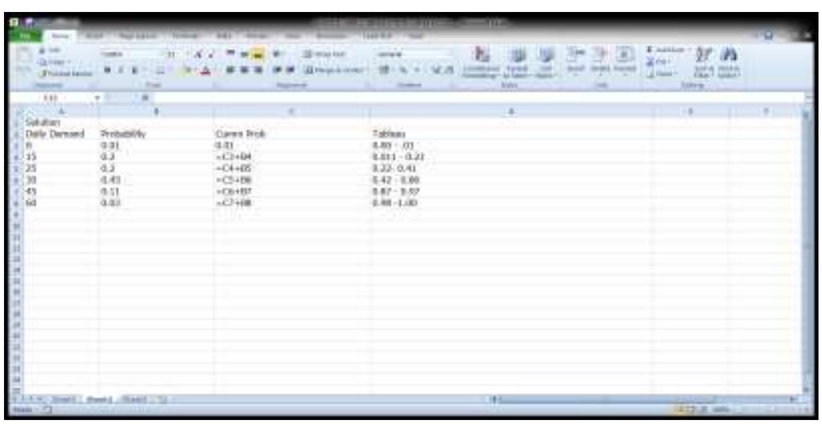
Example of Monte Carlo SimulationUsing *Excel*

A shop keeper wants to maintain stock of a branded product. Previous records show the daily demand pattern of the item with its probabilities given as below. Generate and use random numbers to simulate the demand for 10 days. Find the average daily demand per day.

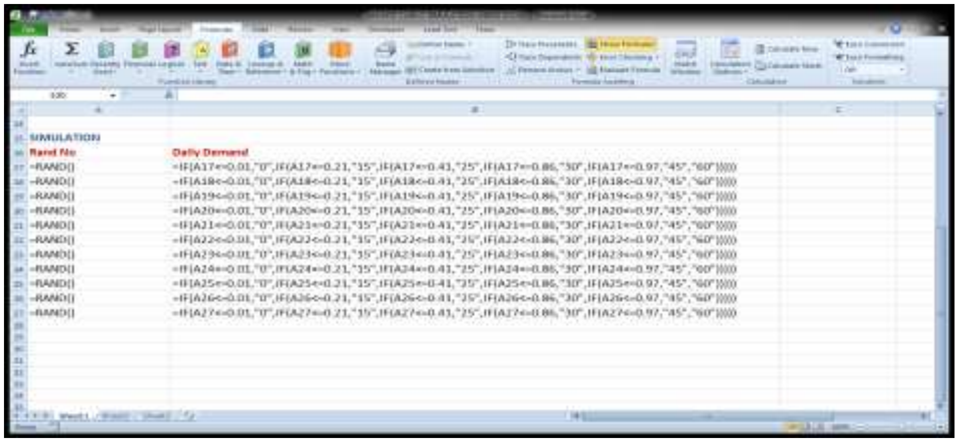
Daily Demand	0	15	25	30	45	60
Probability	0.01	0.2	0.2	0.45	0.11	0.03

Solution Steps

1. **Set up the Tableau**



2. Create a Table of Random Number and program the Excel for Simulation



Final Solution